ME 699 CubeSat Project

Spring 2025 Final Report

Figure 1: Mission Logo

05-14-25

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Introduction

The Precision Hyperspectral Agricultural Tracker Satellite (PHAT-Sat) is a 3U CubeSat designed by an interdisciplinary team of students from Kansas State University's Manhattan and Salina campuses. The mission aims to enhance agricultural sustainability across the central Midwest by using hyperspectral imaging to detect crop health anomalies with improved spectral resolution and revisit rates. PHAT-Sat's goal is to deliver frequent and accurate vegetation index data to support precision agriculture, drought monitoring, and resource management in the Midwest.

This semester, the team researched mission ideas, selected a mission, and engaged in preliminary design of the overall system and subsystems. The final selected mission also incorporated two hosted payloads: a neutron detector built by K-State's nuclear engineering department and thermoelectric generators (TEGs). Extensive research was conducted to identify hardware for each subsystem and payload and requirements were established at the mission, system and subsystem level. The team developed a high-level satellite block diagram, subsystem interface definitions, and mission success criteria.

PHAT-Sat has completed and passed both the Merit and Feasibility Review milestones through the NASA CubeSat Launch Initiative (CSLI) preparation pathway. Both reviews received feedback from several aerospace industry professionals and academics. The project involved 25 students across six engineering disciplines and will continue as a interdisciplinary engineering design course for several semesters.

Listed below are the students along with their majors:

Table 1: Student Major(s) & Minors

Student Name	**Major(s) & Minors**	**Student Name**	**Major(s) & Minors**
Braden Adams	CS & Leadership Studies	Ben Monday	ME
Will Beemer	ME	Chuck Norris	MET
Michael Bennett	Electrical Engineering	Quan T. Pham	ME & Leadership Studies
Jude Conley	Computer Engineering	Nalen Rangarajan	CS & Mathematics
Connor Davis	MET	Derek Rephlo	ME & Management
Nolan Elder	ME	Adam Riekeman	ME & CS
Logan Gamble	ME	Drew Sallman	ME
Jake Hajduch	ME	Phillip Shirkey	ME & Mathematics
Logan Hayward	CS & Mathematics	Anish Srivastava	ME & MBA
Trenton Herrman	ME & Physics	Sawyer Stoskopf	MET
Colby Johnston	ME, NE, & Music minor	Bailey Walke	ME & Physics
Paul Kessinger	ME	Bryce Zavadil	ME
Kyler McKibban	Engineering Technology		

Mission Overview

PHAT-Sat addresses a growing need for more responsive and higher-resolution agricultural monitoring in the United States Midwest. This region produces a significant percentage of the world's corn, soybeans, and wheat. Current satellites, including multispectral satellites like Landsat-8/9 and Sentinel-2, have revisit periods of approximately 16 days and lack the spectral precision needed for early detection of drought, disease, and vegetation stress. Drones and UAVs do not provide consistent data and only offer four to five spectral bands.

The mission's purpose is to design, build, test and launch a CubeSat equipped with a hyperspectral imager to monitor vegetation indices, enabling proactive decisions in precision agriculture. The imager will provide improved spectral information with comparable spatial resolution to current NASA satellites, allowing PHAT-Sat's data to be used to supplement existing datasets.

Mission objectives include:

- Acquiring hyperspectral imagery of the Midwest in 12–40 spectral bands from 450–1200 nm (VNIR), with a target ground resolution of 50m.
- Demonstrating hosted payloads: a miniaturized neutron detector and thermoelectric generator (TEG) to prove their space-worthiness.
- Maintaining mission operability for a minimum of 24 months in a 500 km sun-synchronous orbit.

Mission success will be measured by successful data collection and downlink, ground-validated vegetation indices, hosted payload performance, and a fully sustained active mission life of two years.

Concept of Operations

1. Launch & Deployment
    - PHAT-Sat will be launched to a 500km sun-synchronous orbit on a rideshare through NASA's CubeSat Launch Initiative.
    - Deployment from the dispenser will be followed by automatic detumbling and initial health checks.
2. Commissioning
    - Upon separation, PHAT-Sat will enter a safe mode for system boot-up, health checks, solar panel and antenna deployment, and initial communication link establishment via UHF.
    - Once stabilized all subsystems will undergo calibration and testing.

3. Nominal Operations (Day in the Life)
   - The spacecraft will enter a regular operational loop based on orbital passes:
     - Imaging Passes: When over the region of interest (Kansas, Missouri, Nebraska, Iowa, Oklahoma, and eastern Colorado), the satellite will orient nadir using ADC and activate the hyperspectral payload.
     - Downlink: When in contact with designated ground stations (Figure 3), science data will be downlinked via X-Band, while telemetry is sent via UHF.
4. Extended Mission & Decommissioning
   - After 24 months, PHAT-Sat may continue to operate as power and system health allow. Ultimately, it will passivate and deorbit within five years per NASA guidelines.

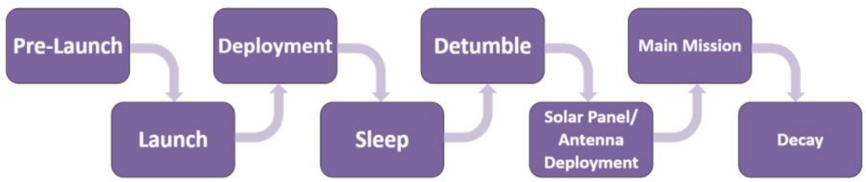


Figure 2: PHAT-Sat Mission Life

Orbit

Phat-Sat will aim for an orbital altitude of 500km with bounds of 475km to 515km. Within this range, the pointing strategy of Phat-Sat can be changed to adjust atmospheric drag to keep PHAT-Sat withing a 2-5 year mission life. Importantly, the 500km orbital altitude range is popular among missions that require sun synchronous orbits, so it is believed that the CSLI and NASA will provide an adequate launch opportunity.

A sun synchronous orbit ensures that Phat-Sat can always image the area of interest during the daylight. An inclination angle of 97.4 degrees ensures that the orbit precesses at the same rate that the earth orbits around the sun, which will keep PHAT-Sat's LTAN constant. An LTAN of between 10:00 and 15:00 is a wide, feasible range allowing extra flexibility. If needed, Phat-Sat can point up to 15 degrees longitudinally to image off nadir (Figure 4).

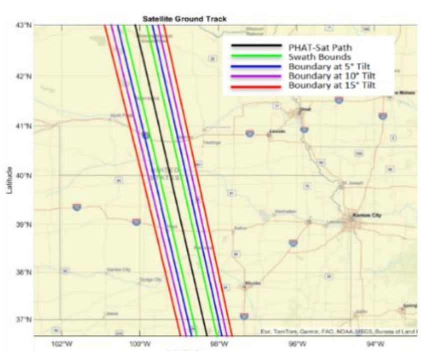


Figure 3: Orbital Path and Swath

Figure 4: Downlink Contact Area

Satellite Overview

PHAT-Sat's primary payload is a hyperspectral imager (1U–1.6U) operating across 12+ bands in the VNIR range (450–1200 nm), producing vegetation indices such as NDVI and EVI onboard. Imaging data is downlinked via a high-speed X-Band transmitter, while command and telemetry functions are handled through UHF. In addition to the core imaging payload, PHAT-Sat includes two hosted technology demonstration units: a miniaturized neutron detector from Radiation Detection Technologies and a thermoelectric generator (TEG) mounted on the rear of the deployable solar panels.

Key subsystems include:

- ADC for precision pointing using reaction wheels, sun sensors, and GNSS.
- EPD for regulated battery and solar power distribution.
- CDH with onboard storage and communications interfacing.
- FSW developed in NASA's Core Flight System (CFS) for autonomous operation.
- Structures designed for launch survivability and deployment compliance.
- Thermal management with heaters, insulation, and heat sinks.

Ground station operations multiple antennas and control systems to receive telemetry, issue commands, and process downlinked science data.

Subsystem layout and trade studies were guided by isometric CAD modeling and are illustrated in the satellite system block diagram (Figure 5). PHAT-Sat will be fully compliant with a selected 3U launch interface deployment envelope.

PHAT-Sat Block Diagram

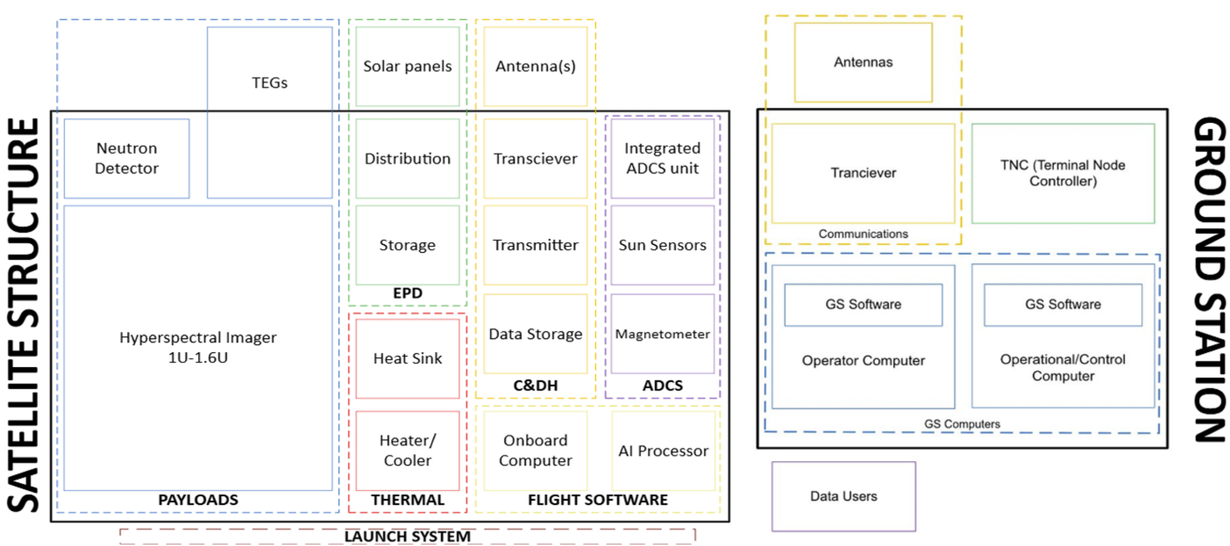
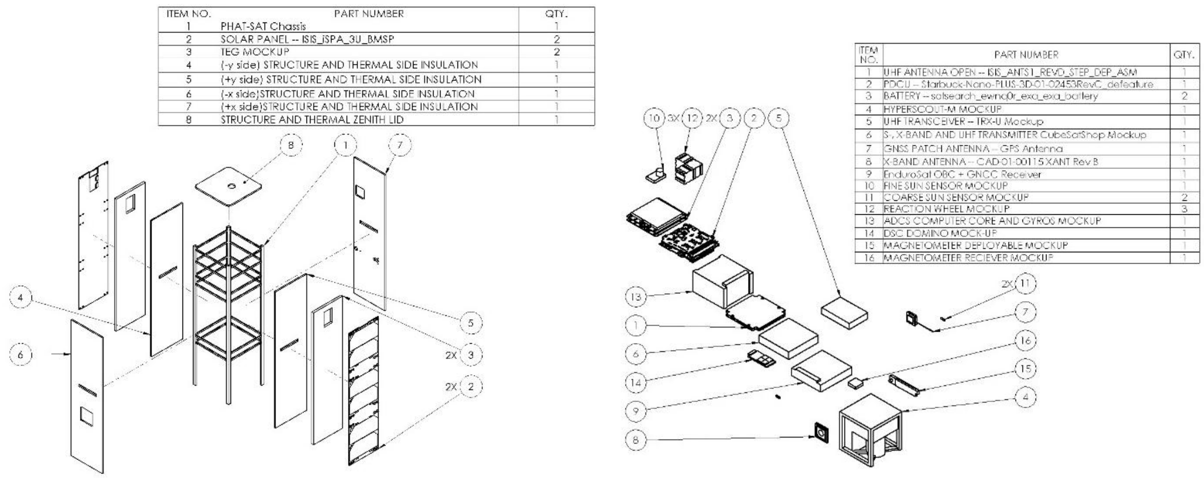


Figure 5: PHAT-Sat System Block Diagram

Isometric Satellite Views



ITEM NO.	PART NUMBER	QTY.
1	PHAT-SAT Chassis	1
2	SOLAR PANEL -- ISIS_iSPA_3U_BMSP	2
3	TEG MOCKUP	2
4	(-y side) STRUCTURE AND THERMAL SIDE INSULATION	1
5	(+y side) STRUCTURE AND THERMAL SIDE INSULATION	1
6	(-x side)STRUCTURE AND THERMAL SIDE INSULATION	1
7	(+x side)STRUCTURE AND THERMAL SIDE INSULATION	1
8	STRUCTURE AND THERMAL ZENITH LID	1

ITEM NO.	PART NUMBER	QTY.
1	UHF ANTENNA OPEN -- ISIS_ANTS1_REVD_STEP_DEP_ASM	1
2	PDCU -- Starbuck-Nano-PLUS-3D-01-02453RevC_defeature	1
3	BATTERY -- satsearch_ewnq0r_exa_exa_battery	2
4	HYPERSCOUT-M MOCKUP	1
5	UHF TRANSCEIVER -- TRX-U Mockup	1
6	S-, X-BAND AND UHF TRANSMITTER CubeSatShop Mockup	1
7	GNSS PATCH ANTENNA -- GPS Antenna	1
8	X-BAND ANTENNA -- CAD-01-00115 XANT Rev B	1
9	EnduroSat OBC + GNCC Receiver	1
10	FINE SUN SENSOR MOCKUP	1
11	COARSE SUN SENSOR MOCKUP	2
12	REACTION WHEEL MOCKUP	3
13	ADCS COMPUTER CORE AND GYROS MOCKUP	1
14	DSC DOMINO MOCK-UP	1
15	MAGNETOMETER DEPLOYABLE MOCKUP	1
16	MAGNETOMETER RECIEVER MOCKUP	1

Figure 6: Exploded and Labeled External and Internal Satellite Components

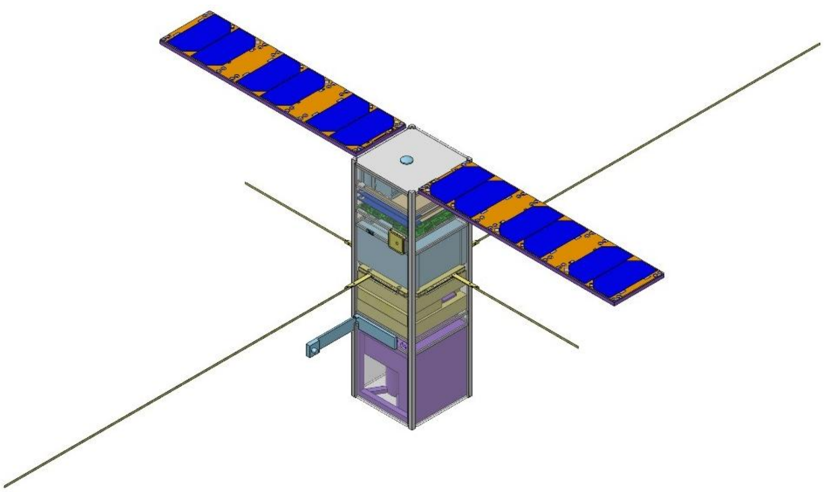


Figure 7: Extended Full Assembly

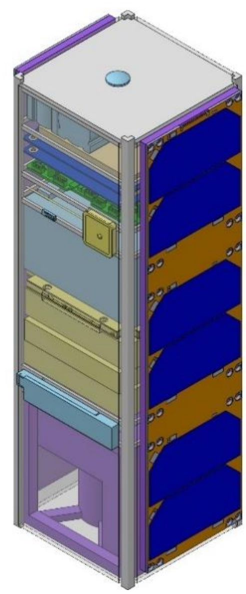


Figure 8: Stowed Full Assembly

Modes and States

PHAT-Sat utilizes operational modes to optimize performance and conserve power. Imaging will only take place over the Midwest, and imaging and downlinking functions will not occur simultaneously to mitigate failure risk and minimize power draw. There are scheduled modes and reactionary/prompted modes based on their function.

Table 2: PHAT-Sat Operational Mode Overview

Mode	Type	Function
Deployment	Scheduled	Initial state post-ejection. Detumbling, solar panel/antenna deployment, and satellite health transmission.
Safe	Scheduled/ Reactionary	Minimal power state for fault recovery and initial startup. Only essential systems remain active.
Nominal	Scheduled	Full operational state. Default mode for health monitoring, transitions to imaging or downlink.
Pointing	Scheduled	ADC aligns satellite for imaging or downlinking modes. Imaging: points imager to nadir. Downlink: points antenna to nadir
Imaging	Scheduled	Hyperspectral imager collects raw data.
Downlink	Scheduled	Antennas align with ground stations. Science and telemetry data transmitted via X-Band/UHF.
Idle	Scheduled	Low-activity state between operations. Conserves power while charging via solar panels and preps for next pass.
Calibration	Scheduled	Used post-deployment and periodically to recalibrate sensors, ADC, and imaging payloads.
Survival	Reactionary	Emergency fallback with strict power conservation.
Receiving	Reactionary	Satellite receives uplink information from ground station.
Sleep	Reactionary	Ultra-low power state for scheduled downtime or eclipse periods.

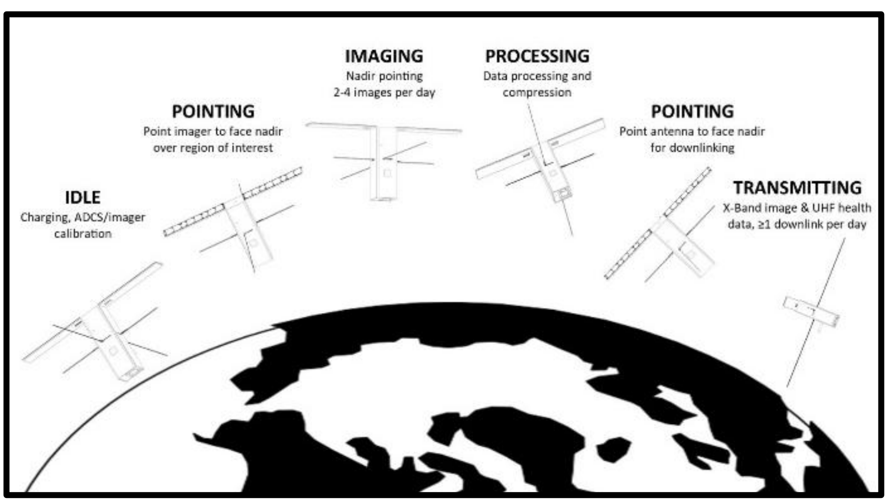


Figure 9: PHAT-Sat "Day in the Life" Diagram

Payload Overview

The payload of PHAT-Sat contains three subsystems: the hyperspectral imager, neutron detector, and thermoelectric generator. These payloads are designed to operate independently, so that scientific and technical missions can be conducted for the duration of the satellite's life, even in a degraded operational mode. The hyperspectral imager is the primary payload and the driving objective of the satellite, and the other payloads are hosted subsystems.

The hyperspectral imager is designed to capture visual and near-infrared imaging data for use in precision agricultural modeling by public and private interests. The payload will image a region of the American Midwest to collect spectral data with a revisit time of 16 days, which will be integrated into existing public data sets (such as those from Landsat-8/9). The data will provide a better time-dependent picture of agricultural trends, which will give the agricultural industry higher awareness of the crop situation during a growing cycle. The data must be publicly accessible, so post-processing and integration with existing public software is required. The satellite must have operational modes to prepare and operate the imager during the sunlit phase of its orbit.

The hosted neutron detector is designed to track neutrons generated by radiation, which pose a hazard to personnel and equipment operating in space. Current detectors have room for improvement regarding space requirements and time response, so testing and space-qualifying a domestic neutron detector will provide K-State and the industry with new capabilities. The data must be accessible to K-State nuclear researchers, so the data collection component of the communications subsystem must be designed to support future detailed modeling done by K-State. The satellite must have operational modes to prepare and operate the detector, with power needs that do not inhibit the imaging mission.

The hosted thermoelectric generator is designed to generate power underneath the same surface area as the installed solar panels by using the available deployment envelope clearance. Power is a serious constraint on satellite design, and solar power is the primary energy source for satellite missions. Testing and space-qualifying a domestic thermoelectric generator will provide K-State and the CubeSat industry with new capabilities. The data must be accessible on request and demonstrate the efficacy of applied thermoelectric generator, so the wattage input of the satellite must be precisely assessed. The satellite must have operational modes to generate power with and without the thermoelectric generator active, to provide test data without inhibiting the imaging mission.

Hyperspectral Imager

The hyperspectral imager is capable of collecting detailed hyperspectral images of the Earth in the visible near-infrared spectrum. The imager is comprised of a camera body, lens, and electronics. The Cosine HyperScout M is the team's primary imager option, as shown in Figure 2 below. The HyperScout M is a compact 1U hyperspectral imaging system that comes integration ready. A secondary imager option is the Raptor Photonics Owl 2560, as shown in Figure 3, which comes as a standalone camera body with an optional lens.

Figure 10: HyperScout Cosine M [1]

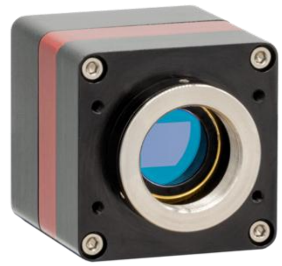


Figure 11: Owl 2560 Camera [2]

The camera body houses the camera sensor, which converts photons to electrical signals and creates an image. The lens provides the necessary magnification to image our desired swath and ground sampling distance (GSD). The imager electronics will control the camera and act as a data and control interface for the whole system. The GSD and swath are driving requirements for this mission, both of which are dependent on the imager sensor and lens. It is key to select an imager that balances the product specifications to meet the requirements of our mission. Our mission requires a GSD of 50m or less and a swath 170km or greater. To meet these requirements, a large-format sensor with a small pixel size is ideal. A large sensor will increase our swath to the desired 170km and a small pixel size will result in a higher resolution image. Table 2 below shows the swath and GSD calculations for the HyperScout M and Owl 2560.

Table 3: Cosine HyperScout M Calculations:

Cosine HyperScout M		
Pixel Size	5.5	um
Sensor Width	22.5	mm
Focal Length	50	mm
Orbit Altitude	500	km
Swath	**225**	**km**
GSD	**55**	**m**

Table 4: Owl 2560 Imager Calculations

Raptor Photonics Owl 2560 w/Custom Lens		
Pixel Size	3.45	um
Sensor Width	8.85	mm
Focal Length	35	mm
Orbit Altitude	500	km
Swath	**126**	**km**
GSD	**49**	**m**

Neutron Detector

The hosted payload features a compact neutron detector developed by Kansas State University's Radiation Detection Technologies (RDT). It includes a built-in logic board for storing and outputting accumulated neutron counts and supports both thermal and fast neutron detection. Built-in background radiation compensation ensures accurate performance in high gamma-flux environments, while an onboard temperature sensor helps monitor conditions across its –20 to 55 °C operating range.

Designed for efficiency and scalability, the detector consumes less than 1 mW and costs around $1,000. The 1-D tileable amplifier IC module provides digital TTL output for easy CubeSat integration. Despite its compact size (38 × 27 × 5 mm) and weight (9.5 g), it offers 30% ± 1% thermal neutron efficiency, <1 count/min gamma sensitivity, and covers a 4 cm^2 area, making it ideal for neutron monitoring in space-constrained applications.

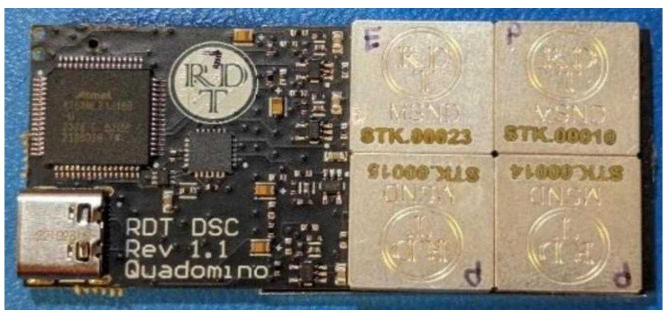


Figure 12: RDT's Domino neutron detector

Thermoelectric Generator

The objective of this hosted payload is to demonstrate CubeSat scale thermoelectric generation technology while generating additional power for the mission. The TEG is produced from bismuth telluride p-type or n-type ingots or similar material. The TEG is intended to extend about 4mm in thickness from the rear surface of the photovoltaic array. The deployment envelope allows TEG sufficient clearance. Extruding the PV array for TEG will require a modification to the panel hinge. We desire a ΔT of 80+ °C. The hot side conducts heat from the PV panel. The cold side of TEG would feature a thin cold sink that radiates heat to space when the sun is normal to the PV array.

TEG costs approximately $24k to purchase or $8k to produce in-house for a full 3U array. Due to mass restrictions, the final size is likely to be less. In-house production would require additional equipment. There are companies that specialize in custom TEG production. Notably, TECTEG Manf. offers a range of TEG substrates to use.

Ground Overview

Ground System Description

A satellite ground system is a critical infrastructure that supports the operation of satellites throughout their lifecycle—launch, early orbit phase, mission operations, and decommissioning. It includes all hardware, software, and personnel required to monitor, command, control, and receive data from the satellite.

The satellite ground system is composed of several integrated components that function together to ensure successful communication, control, and data management for an orbiting satellite. At the center of this system is the Mission Operations Center (MOC), which serves as the central hub for planning and overseeing satellite missions. Supporting the MOC is the Ground Control Center (GCC), which handles the real-time operational control of the satellite. It serves as a critical link between the planning activities of the MOC and the physical transmission infrastructure of the ground stations.

The Ground Antenna System, often referred to as the ground station, includes antennas and radio frequency (RF) hardware capable of tracking satellites as they pass overhead. These stations are equipped to transmit uplink signals (commands) and receive downlink signals (telemetry and payload data).

Ground System Block Diagram

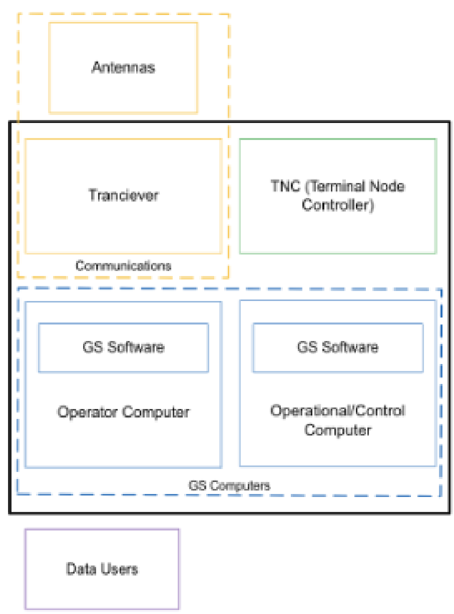


Figure 13: Ground Station Block Diagram

Ground Station Description

The architecture of a CubeSat ground station can be divided into several major subsystems: the antenna system, the radio frequency (RF) subsystem, the baseband and signal processing unit, tracking and control software, and the data/network management system [3].

The antenna system typically consists of one or more Yagi-Uda antennas for very high frequency (VHF) and ultra-high frequency (UHF) bands, and small parabolic dishes for S-band communication. Yagi antennas are favored for their affordability, ease of construction, and sufficient gain for low-data-rate CubeSat missions [4]. In contrast, small dishes are used when higher gain is required, particularly for S-band payload downlinks [4]. Antenna mounts generally use an azimuth-elevation configuration, enabling them to track satellites throughout their passes. Manual tracking is feasible for educational or low-budget missions, but automated rotators improve accuracy and reduce labor.

The RF subsystem includes low-noise amplifiers (LNAs) to improve the signal-to-noise ratio during downlink, power amplifiers for uplink transmissions, and various RF filters to suppress interference. This subsystem is also responsible for frequency conversion, which adjusts the satellite's transmitted frequency to an intermediate or baseband frequency for processing [4]. A TNC module can be used for the modulation and demodulation of data packets for it to be interpreted by the computers. The TNC module works with the transceiver and ground computer to achieve accurate and efficient data communication.

Finally, the data and network management component ensure that mission data is stored securely and made accessible to stakeholders.[1] This system may include a local server or a cloud-based infrastructure, as well as remote access tools such as SSH or VPN to support distributed operation.

Ground station location is an important aspect of the overall operation. Locations need to be selected based on PHAT-Sats passes while allowing low interference data communication. Current locations being explored include CU Boulder, Morehead State, and K-State. K-State radio club’s current equipment would allow for VHF/UHF communications, and they are very interested in upgrading equipment to allow S and X band communications. CU Boulder and Morehead State’s potential involvement is still underway. Other options to be explored include the U.S. Naval academy, U.S. Airforce Academy, Kongsberg Satellite Services, Swedish Space Corporation, and AWS Ground Station.

Trade Studies for CubeSat Ground Stations

Several trade studies are essential during the ground station design process to ensure the system meets mission requirements while staying within budgetary and technical constraints.

One of the primary trade-offs is between antenna types. Yagi antennas are inexpensive, easy to deploy, and ideal for VHF/UHF frequencies, making them suitable for basic telemetry and control [4]. However, for higher data rate applications, a small parabolic dish may be necessary, particularly in the S-band. Dishes offer higher gain but at the cost of increased complexity and reduced portability.

Another important decision involves the rotator system. Manual tracking systems offer a low-cost solution, especially for short-duration missions or educational use. However, automated tracking systems, while more expensive, significantly reduce operator workload and improve reliability during critical communication windows.

Frequency selection also plays a major role in ground station design. UHF/VHF bands are commonly used for telemetry, tracking, and control due to their resilience and simpler licensing under amateur radio regulations [3]. However, they offer limited data rates, typically under 10 kbps. For payload data requiring higher throughput, the S-band provides a reasonable compromise between bandwidth and complexity, albeit with stricter licensing requirements and greater susceptibility to weather-related signal degradation [4].

In the area of baseband processing, mission planners often choose between software-defined radios (SDR) and commercial hardware modems [3]. SDRs are attractive due to their low cost, adaptability, and strong support from amateur radio and academic communities. However, they come with a steeper learning curve and may require more engineering effort to develop reliable decoding chains. Conversely, commercial modems offer plug-and-play reliability at a higher cost and with less flexibility.

Ground Software Description

The purpose of the ground software is to handle the telemetry from the ground station to the satellite. To do so, we plan to use an open-source software such as Cosmos from Ball Aerospace [3]. This software will need to send commands, receive files, and check the heartbeat of the satellite. Cosmos offers these features as “out of the box functionality [4]. Along with these features, Cosmos offers a testing suite capable of communicating with any hardware that has a defined communication protocol. The majority of the work with commands and downlinks lies in the future. While we have selected our platform, the youth of the project has not allowed us to test this software with an actual satellite. More details will be provided at that point.

Ground Data Description

One of the main deliverables of this mission is a widely distributed data network. To achieve this goal, our ground software team met with Dr. Shawn Hutchinson to discuss how the images we retrieve can be distributed using current infrastructure in place at K-State. Through this conversation, we learned about the use of an Amazon Simple Storage Service (S3) bucket and the ESRI Web Portal to store and distribute images respectively. Each image we receive from the satellite will contain information on the imaged area from each wavelength we choose. This will all be able to be stored in the S3 bucket as a mosaic dataset, a common format for GIS. From that point, stakeholders will be able to request wavelengths or combinations of wavelengths from our ground station in order to receive images that are basically made to order. This approach not only allows users to view very personalized images, but our team will be able to verify our images compared to those of larger satellites within ESRI. Combining these functionalities in our distribution system delivers the main purpose of this mission, data that is both easy to access and comparable to larger missions. The work to make this happen is set to begin in the fall, more details will follow at that time. An early diagram of the distribution system is available below.

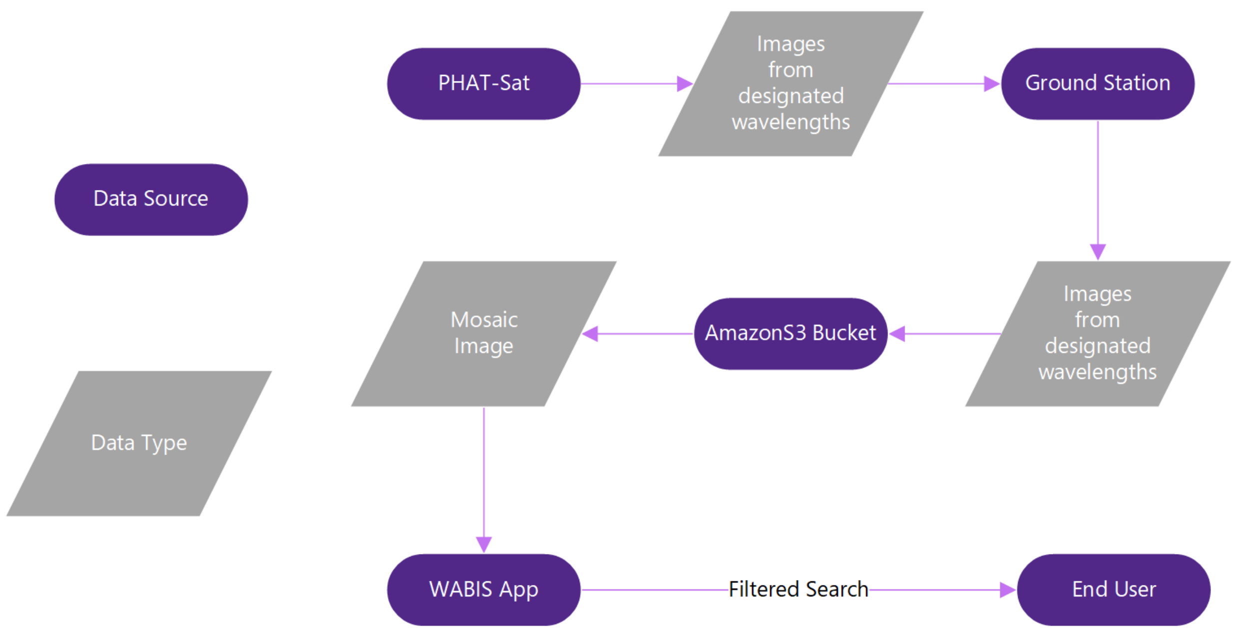


Figure 14: Data Distribution Diagram

Ground Support Equipment Description

Based off information from a master's thesis at MIT titled "*A Systems-Engineering Assessment of Multiple CubeSat Build Approaches by Zachary Scott Decker*". To define a basis for mechanical ground station equipment (mechanical GSE): (Components and/or equipment which is not part of the payload, is considered to be an extension of GSE. Thus, any non-payload related equipment that is structural and not electrical, may be considered as mechanical GSE.) Examples of this are items such as integration stands, carrying cases and testing equipment.

Items we specifically would use are integration/testing stands and a carrying case. To keep the satellite from being damaged, using a hardshell carrying case such as a Pelican iM2200 Storm Case would prevent the satellite from being damaged during transportation. Possibly even the manufacturing of an aluminum mounting system with handles that sat inside the case would allow for a more secure method of removing the payload from the case. Furthermore, the use of an integration/testing stand that the proposed aluminum mount could fasten too, would allow for work to be done on the payload with ease, and a more stable environment. As well as mounting the payload to vibration test equipment to verify the device would withstand the delivery method.

The Electrical Ground Support Equipment (EGSE) for PHAT-Sat is essential for safe, repeatable integration and functional testing of the spacecraft's electrical subsystems, including its hyperspectral imaging payload. EGSE communication with PHAT-Sat's onboard computer (OBC) will be established via a USB-to-UART or USB-to-RS-422 adapter, enabling real-time command and telemetry exchange. A development laptop at K-State facilities will serve as the ground station interface, running either custom scripts or open-source ground software like COSMOS. Arizona State University and Georgia Institute of Technology have used similar software in their CubeSat's such as REXIS (Georgia Institute of Technology). This interface will be essential for verifying operational modes including imaging and downlink preparation. Power for integration and subsystem testing will be supplied by a programmable bench-top DC power supply (ex. BK Precision 9201B), capable of simulating the spacecraft's power conditions during orbital operations. An inline current/voltage will allow real-time tracking of power demands, especially during hyperspectral imaging, which requires elevated current draw. These tools will be used to evaluate performance against subsystem requirements and verify that load margins are within flight tolerances. To test the VNIR hyperspectral imager's functionality and interface, the EGSE will include a USB 3.0 frame grabber and software tools such as ImageJ, MATLAB, or Python-based spectral analysis tools. This will allow ground teams to trigger the sensor, and verify calibration steps such as band alignment, NDVI accuracy, and thermal loading.

In summary, the EGSE system for PHAT-Sat will provide a flexible, standardized, and scalable approach to electrical integration and testing. Its design draws on practices from other academic CubeSat programs while addressing PHAT-Sat's specific goals: validating a multi-sensor platform for agricultural remote sensing and preparing for a robust, multi-year mission.

Bus Overview

The bus's primary purpose is to host the payloads, and the competing needs (of the bus and payload) must be balanced. As a high-level overview of bus interfaces, the qualitative needs of each system from the others are listed below. This gives a general framework for what connections are needed to ensure satellite requirements are met.

- *Payloads (Imager, TEG, Neutron Detector)*
    - Accurate nadir pointing and image stabilization data (ADC—mainly TEG and Imager)
    - data storage, handling, and downlinking (CDH and FSW)
    - sufficient energy (EPD)
    - secure and vibrationally safe securing (Structure); maintaining proper temperature bound (thermal)
- *Attitude Determination and Control (ADC) Needs*
    - data storage, handling, and downlinking (CDH and FSW)
    - sufficient energy (EPD)
    - secure and vibrationally safe securing (Structure)
    - maintaining proper temperature bound (thermal)
- *Communications and Data Handling (CDH) Needs*
    - Payload status data and measurement (Payloads)
    - Bus subsystem status data (ALL subsystems)
    - sufficient energy (EPD);
    - secure and vibrationally safe securing (Structure)
    - maintaining proper temperature bound (thermal)
- *Electrical Power and Distribution (EPD) Needs*
    - secure and vibrationally safe securing (Structure)
    - maintaining proper temperature bound (thermal)
    - Power produced by solar arrays and TEGs (Payloads)
- *Flight Software (FSW) Needs*
    - Ultimate control over satellite/all subsystems (ALL)
    - Data access, interpretation, and response (CDH)
- *Structure Needs*
    - Proper integration into the launch system
    - Satellite layout for fastener locations (ALL)
- *Thermal Needs*
    - Sufficient energy (EPD)
    - Satellite layout for proper heat management (ALL)

Attitude Determination and Control

The attitude determination and control (ADC) subsystem points the satellite to targets. Gyroscopes and magnetometers are for attitude determination. Reaction wheels and magnetorquers are for attitude control. The satellite navigation system is for position determination. Sun sensors are for sun tracking in power collection and protecting the imager from direct sunlight. The subsystem will require a computer interface. Performance characteristics will be defined once the satellite structure and mass are fixed. Values for accuracy and precision are derived using the smallest applicable constraint from an imager resolution of 50 m, orbital velocity of 7.616 km/s, and imager exposure time of 1 s.

Table 5: ADC Subsystem Pointing Budget

Subsystem	Component	Target	Axis to Align and Frame of Axis	Accuracy (+,-)	Precision (+,-)	Function
IMG	Camera	North America	(+) Z Axis of Spacecraft to be aligned with Nadir (Earth)	0.0057 0.064/s	0.0057 0.0057/s	Imaging Mission
CDH	Antenna	Ground Station	(+,-) X or Y Axis of Spacecraft to be aligned with Nadir (Earth)	5 1/s	5 1/s	Data Downlinking
EPD	Solar Panels	Sun	(-) Z Axis of Spacecraft to be aligned with Zenith (Sun)	5 1/s	5 1/s	Power Collection
ADC	GNSS Receiver	NA	(+,-) X or Y Axis of Spacecraft to be aligned with Nadir (Earth)	50m	50m	Position Determination
	Inertial Measurement Unit (GYRO+MTM)	NA	NA	0.0057* 0.064/s*	0.0057* 0.0057/s*	Attitude Determination
	Fine Sun Sensor	NA	(-) Z Axis of Spacecraft to be aligned with Zenith (Sun)	5	5	Attitude Determination, Sun Tracking
	Coarse Sun Sensor	NA	(+) Z Axis of Spacecraft must point away from Zenith (Sun)	NA	NA	Sun Avoidance
	Reaction Wheel	NA	NA	NA	NA	Attitude Control
	Magnetorquer	NA	NA	NA	NA	Attitude Control
***GND**	Ground Processing	NA	NA	0.0057	0.0057	Position Determination

*If ADC cannot reach the required accuracy or precision during imaging, more ground post-processing will be needed.

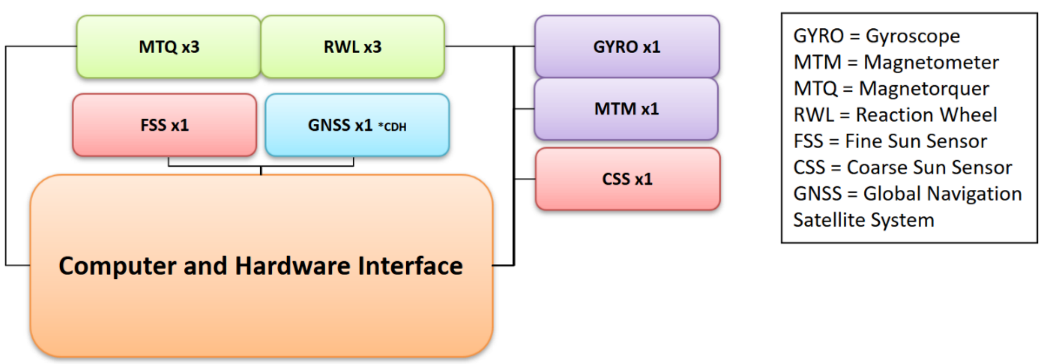


Figure 15: ADC Subsystem Block Diagram

Gyroscope (GYRO)

Gyroscopes measure rotation and rate of rotation on a given axis or axes. This is used for attitude determination, and to define precise attitude corrections during imaging. It is also used for detumbling, where reducing the satellite's rate of rotation is required. Precision gyroscopes are necessary for ADC, but they should not be used to control orientation (such as with a control moment gyroscope) due to the restricted nature of the technology.

Magnetometer (MTM)

The magnetometer measures the direction of the Earth's local magnetic field relative to the orientation of the satellite. The vector corresponding to this direction describes the orientation of the magnetometer in space, so the angular position can be determined.

Reaction Wheel (RWL)

At a minimum, three reaction wheels will be integrated in a perpendicular arrangement to generate torque on all three axes. More wheels may be used for redundancy. Reaction wheels provide torque when powered, which creates an angular acceleration in the opposite direction for PHAT-Sat. These wheels have a maximum angular velocity, after which they need to desaturate (lower their velocity) in order to continue operating.

Magnetorquer (MTQ)

Magnetorquers generate magnetic moments by running current through a coil. This moment interacts with the Earth's magnetic field to generate torque. They can be used to apply an equal and opposite torque to the reaction wheels, desaturating them and allowing PHAT-Sat to maintain orientation and angular velocity. There will be three magnetorquers placed perpendicular to each other, like the reaction wheels.

Fine Sun Sensor (FSS)

Fine Sun Sensors are a high-sensitivity imager designed to be operated in direct sunlight. When the Sun enters the sensor's field of view (FOV), the FSS finds its position. Based on PHAT-Sat's geometry, the solar panels can then be oriented for efficient power collection. Sun tracking during imaging ensures that the imager will not be exposed to direct sunlight.

Coarse Sun Sensor (CSS)

Coarse Sun Sensors detect the presence of sunlight using a photodiode. A CSS will be installed alongside the imager, to protect the imager from direct sunlight by notifying the ADC computer to begin sun avoidance procedures when it is activated.

Global Navigation Satellite System (GNSS)

The satellite navigation system will use GNSS or Global Positioning System (GPS) satellites in higher orbits to find the satellite's position relative to Earth. Because the imaging mission produces data that must be localized in space and time, the GNSS system must provide sufficient knowledge through differential GPS to define an image's ground location within 10 m. Systems that offer position knowledge superior to +/- 5 m are restricted by law.

Communications and Data Handling

Block Diagram

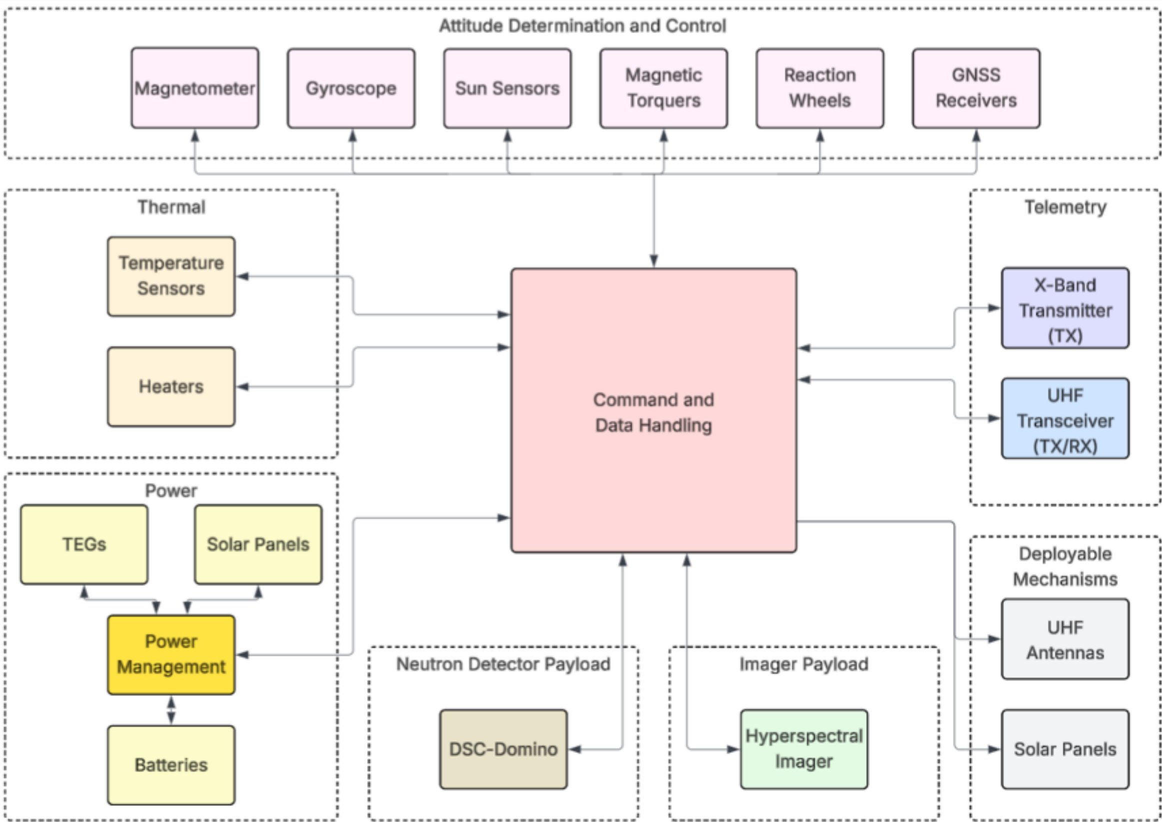


Figure 16: CDH Block Diagram

Data Budget

A data budget [3] provides a way to approximate network use for each component of the satellite. Metrics associated with each component are quantified as ***"high (strict for timing),"*** ***"medium,"*** or ***"low"*** based on estimated usage. Metrics included are ***throughput estimation, CPU utilization, non-volatile memory, volatile memory, update frequency,*** and ***timing accuracy***. Values and cutoffs will become more defined as components are selected and data sizes are solidified. Current cutoff estimates are > 1 Mbps for ***High***, between 1 Mbps and 10 Kbps for ***Medium***, and < 10 Kbps for Low. Page 28 pictures the data budget table where each row corresponds to a satellite component, with each column corresponding to a given metric.

Table 6: CDH Data Budget

Subsystem	Component	Throughput Estimation	CPU Utilization	Non-volatile Memory	Volatile Processor Memory	Update Frequency	Timing Accuracy
Payload	Imager	High	High	High	High	High	Strict
	DSC-Domino	Low	Low	Medium		Medium	Strict
Power	Battery	Low	Low	Low		Medium	Medium
	TEGs	Low	Low	Low		Medium	Strict
	Solar Panels	Low	Low	Low			Medium
FSW	On-Board Computer	Medium	High	Medium	High	High	Strict
CDH	Transceiver	Low	Low	Low			
	Transmitter	High	Medium	High			
ADC	Various Components	Medium				Medium	Strict
Thermal Control	Temp Sensor	Low	Low	Low		Low	Low
	Heaters	Low	Low	Low			

Link Budget

A link budget [4] provides a way to quantify the signal connection between a satellite and the corresponding ground station. The link budget equation [5] to calculate this connection in dBs is comprised of the following:

[Transmitter Power] + [Transmitter Antenna Gain] - [Transmitter Loss] - [Free Space Path Loss] - [Miscellaneous Loss] + [Receiver Antenna Gain] - [Receiver Loss].

The resulting value tells the likelihood of successful communication from satellite to ground station. A value > 0 dBs is considered reliable, while a value <= 0 dBs is considered unreliable.

The current link budget calculations are described in table 7. Since ground stations have not been solidified and information is missing, parameters [Receiver Antenna Gain] and [Receiver Loss] are subject to change. This will drive ground station selection, so it is crucial that ground stations are contacted as soon as possible.

Table 7: CDH Link Budget

Frequency	Received Power
X-Band Tx (~8000 MHz)	-127 + [Receiver Antenna Gain] - [Receiver Loss] *need ground station information to complete
UHF Tx/Rx (~438 MHz)	-136.5 + [Receiver Antenna Gain] - [Receiver Loss] *need ground station information to complete

Electrical Power and Distribution

The electrical power and distribution (EPD) subsystem provides power generation, storage, control, and distribution to the vehicle. The main components of the EPD subsystem are the photovoltaic arrays, power conditioning and distribution unit (PCDU), and lithium-ion batteries. Power generation is accomplished through the utilization of photovoltaic arrays. The PCDU will control and distribute the accumulated power to appropriate subsystems as needed. The batteries will store the power provided by the photovoltaic arrays and provide power to subsystems during "dark" periods of little to no sunlight power generation. The following block diagram illustrates the EPD subsystem, with red arrows representing power flow:

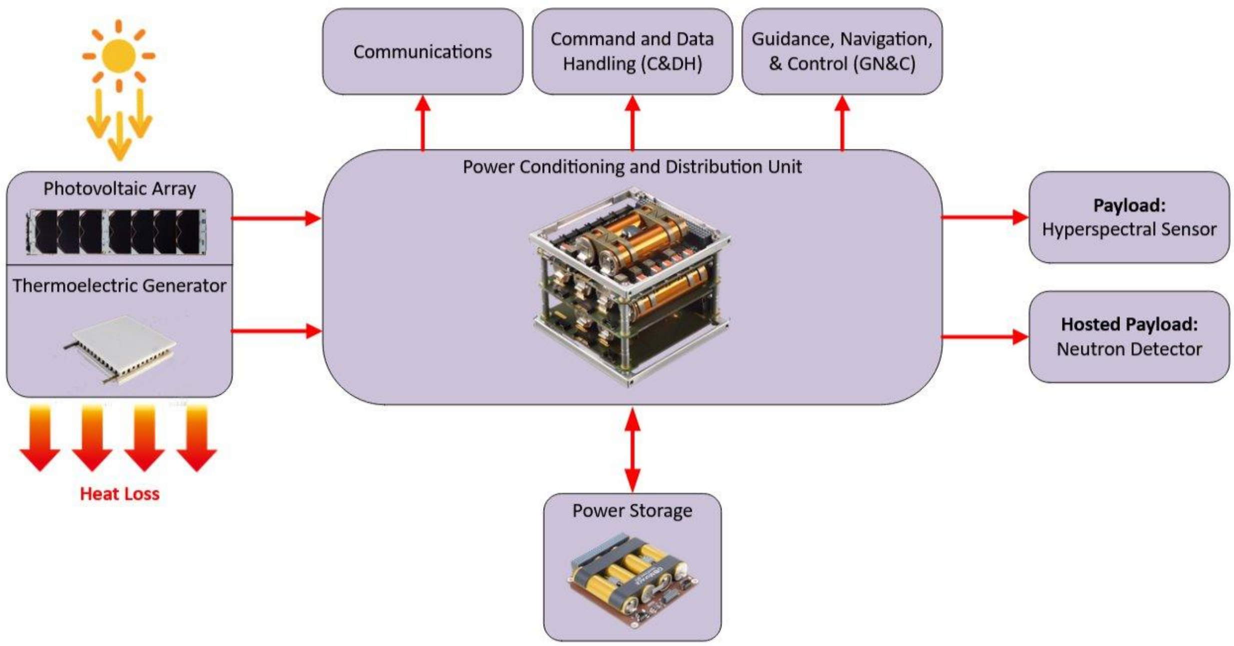


Figure 17: EPD Block Diagram

Overall size and mass will ultimately be based upon vehicle needs, once all subsystems are fleshed out with components selected and a better understanding of orbit schedule and component duty cycles. The photovoltaic arrays will be an estimated 118 – 285 g each and consist of two to three arrays, depending on overall power requirements. Each array will provide an advertised peak of 8W, with an estimate 4.8W average orbit power per array. The PCDU may either be purchased as an assembly including energy storage or possibly designed and built in house by a Phat-Sat team member depending on the direction of the project. The amount of storage required will be a calculated value and kept above a minimum storage amount to avoid the lower battery curve anticipated in design. The goal of the project will be to reduce the storage size to just the right amount of storage to meet mission requirements, while minimizing weight and size of the battery space. The following power budget example will be used to estimate vehicle power requirements over a typical orbit cycle:

Table 8: Power Budget

			Orbit				Eclipse	
			Power	**Num:**	**Duty Cycle:**	**Watts:**	**Duty Cycle:**	**Watts:**
EPD								
	Peak Power	8						
	Orbit Average Power	4.8						
Payload								
	Hyperspectral			1				
	Neutron Detector		0.05	1				
ADCS			1.3					
COMM								
C&DH								
Thermal								

Without parts selected it is not possible at this time to accurately construct a power budget. From the table, we see the important columns needed for a power budget are power, number of devices, duty cycle of device, and watts. The orbit cycle and eclipse cycle will each need to estimate the duty cycle in which the device is active for that period of the orbit. With these values it becomes a mathematical exercise in which power is allotted per device during periods of activation. We will be able to estimate the power needs from this table and decide upon the actual component makeup of each subsystem as well as estimated storage capacity needed to survive eclipse periods.

Thermal

The thermal control system ensures critical components remain within their acceptable temperature ranges, which can differ for active and inactive states throughout the mission. Based on initial heat transfer calculations, irradiance from the sun will result in about 41.6 W of heat transfer to PHAT-Sat while internal heat generation will result in about 4.4 W. For 45-minute periods, we will use the insulation and radiator to mitigate this heat gain. In eclipse, 6 W of heat gain will be received while 4.4 W of internal heat generation occurs.

To keep the satellite from overheating, components will be connected to thermal straps that can conduct heat to a radiator on the cold side. It is likely that passive thermal control (only radiators, insulation and heat straps) will be enough to maintain thermal ranges. Specific components such as the battery may feature built-in resistive heaters. The thermal ranges included in the block diagram are PHAT-SAT alterations on data included in the Critical Design Review for Phoenix CubeSat. PHAT-SAT orientations and the varying thermal conditions will need to be considered to ensure that no systems are damaged.

Block Diagram

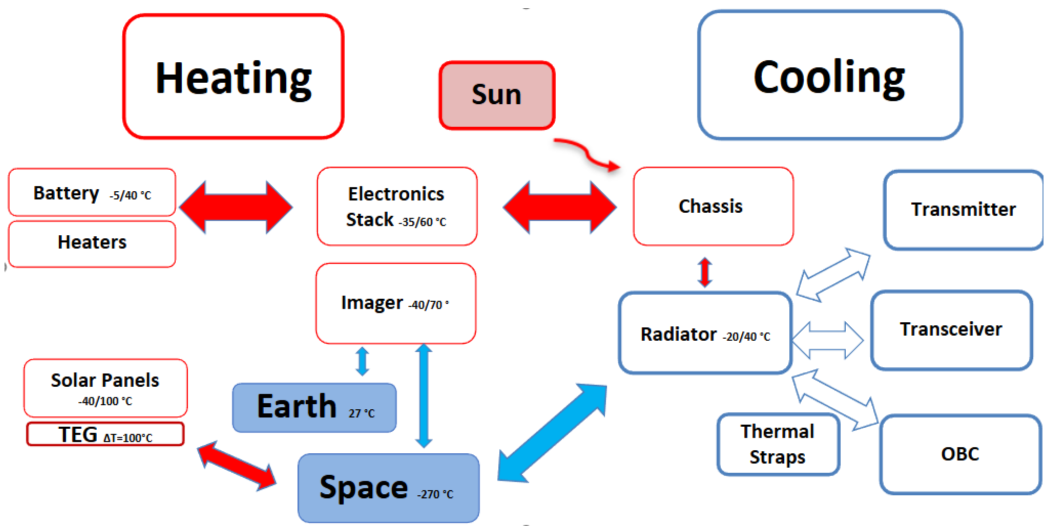


Figure 18: Thermal Block Diagram

Thermal Control Table

The thermal control table is based on Phoenix CubeSat's critical design review documentation [8] it is entirely possible that it will be decided that temperature sensors will not need to be on every component but considering that thermistors are very light-weight and small the burden may lay more on the computational and electrical aspect than physical constraints.

Table 9: Thermal Control Table

Component	Imager	Radiator	Solar	TEG	Batt.	ADC	Transmitter	Receiver	UI Board	OBC	Transceiver	EPD
Thermistor Count	3	1	2	6	5	2	4	1	1	2	2	1
Heater Count					1							

Structure

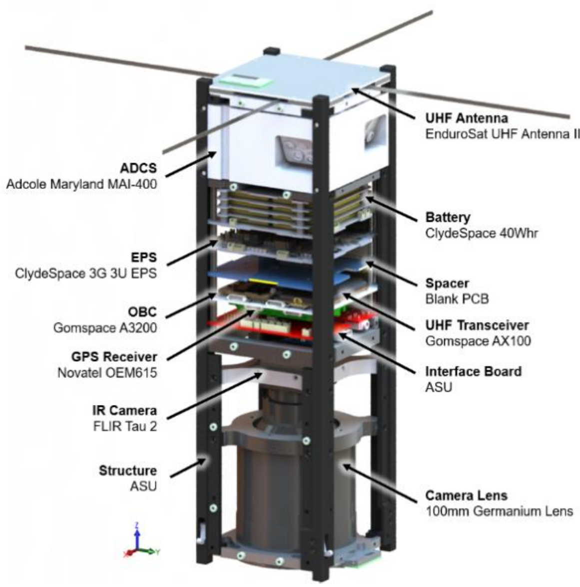


Figure 19 : Structure Diagram for Optical CubeSat [8]

The structure (STR) subsystem provides a chassis to physically contain and mount all components of the satellite. As a 3U satellite, PHAT-SAT will measure 30 cm in the z-direction (towards Earth) and 10 cm in the x- (towards 8orbit velocity) and y- (towards the Sun) directions. The chassis will also be attached to deployment rails that will protrude slightly for the purpose of interfacing during the launch. The location of the hyperspectral imager on PHAT-SAT is crucial to keep the field of view clear during imaging. Additionally, the locations of sensors and antennas will need to be carefully considered for downlinking, thermal, solar and ADC purposes.

Precise alignment of the mounted components and the chassis itself during assembly of PHAT-SAT will be necessary to ensure the accuracy of the hyperspectral images and ADCS sensors. If misalignment does occur there is a possibility the issue could be resolved with software correction.

Block Diagram

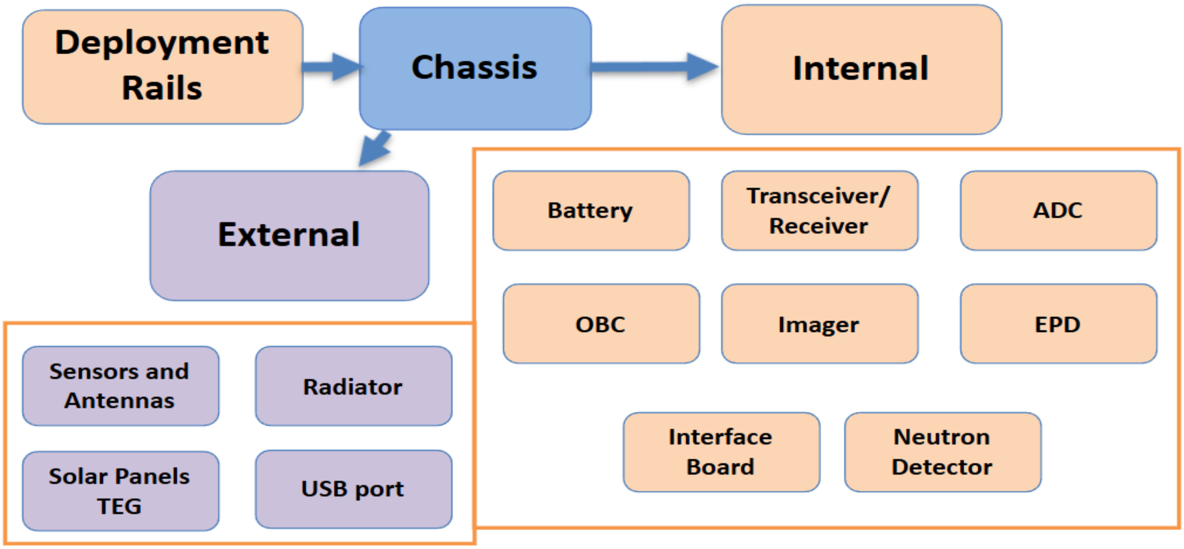


Figure 20: Structures Block Diagram

Mass Budget

The mass budget will be used in making choices for components. With 3U CubeSats generally restricted to a mass of about 3-4 kilograms there will likely be trade-offs and compromises that will be made to keep the satellite's mass within an acceptable range. All values with an asterisk were taken from the PHAT-SAT master equipment list. Other values were extrapolated from the mass budget for a 2U CubeSat [9].

Table 10: Mass Budget

Component	Mass (kg)
Hyperspectral Imager*	1.2
Neutron Detector*	.304
Thermal Straps*	.006
TEG*	.2
PDCU*	.148-.23
ADC*	.575
Battery	.55
Antenna	.05
Transceiver	.04
Neutron Detector*	.015
OBC	.04
Thermal Sensors (Thermistors)	.001 (per unit)
Horizon Sensors	.05-.3 (per unit)
Sun Sensors	.005-.03 (per unit)

Flight Software

Flight Software (FSW) is the central "brain" of the satellite mission. Its primary function is designed to support hyperspectral imaging from LEO for agricultural sciences. The FSW will carry out coordination between subsystems, communication with ground station(s), execution of science objectives, and monitor and log overall system health and events.

The on-board computer (OBC) will be used for hosting the FSW and serves as a hardware link to each subsystem, facilitating networking to send and receive data and commands. Just as the FSW is the "brain" of the satellite, the OBC is the "heart."

Given current mission complexity and limited space for redundancy, the FSW must be robust, reliable, and efficient, with effective error handling capabilities. Furthermore, the FSW will be responsible for carrying out key responsibilities of subsystems like communication and data handling, power distribution, payload control, and more.

Please refer to Appendix C where the Software Development Plan is located, detailing each part of the FSW Development process and its corresponding management plan.

Launch Interface

The launch interface subsystem determines how PHAT-Sat will be secured during launch and identifies a launch provider. The primary strategy is to apply through NASA's CSLI, which offers universities access to space via sponsored launches [10]. PHAT-Sat is designed to for a 3U Rail-Guide dispenser, though the final dispenser and launch vehicle will be determined by NASA if the proposal is selected [10].

To mitigate risk, a backup plan involves using Exolaunch's 3U EXOPod with rail lengths of 340.5 mm. This dispenser offers a strong flight heritage and in-house testing [11]. Rideshare options from companies like SpaceX could meet the mission's orbit deployment requirement of 500 km. This subsystem will continue to assess both CSLI and private commercial launch options, leveraging costs, timelines, and dispenser integration to PHAT-Sat.

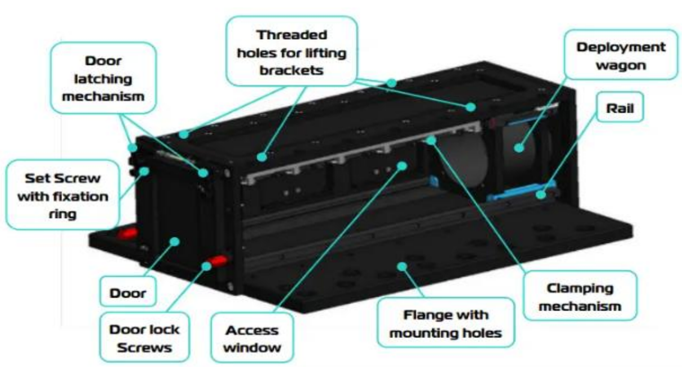


Figure 21: EXOPod Model Features

Management Overview

For the Spring 2025 semester we chose a three-tier organizational structure, the first tier is "the front office" it consists of the PI, PM, Business Operations, and the top leadership team of the CubeSat team. The purpose of this team exists to keep the rest of the team on track and accountable for the purpose of the project. The second tier is the subsystem overview, these are the overarching leads over a particular part of the mission, this tier is all about getting into the day-to-day operations without getting into the weeds of any issue. Weekly updates stem from tier 2 advancements to the general project. Tier 3 is the most productive tier on any day-to-day issue, tier 3 is about sorting out, designing, researching, the on-the-ground work of the project. Issues discussed in tier 3 should not alter the purpose of the mission or operations but rather work towards meeting the operations and missions established by tier 2 and tier 1. In the final hierarchy Business Operations is tier 2 with input at the tier 1 level. Tier 3 business operations would include accounting, marketing, sponsorship outreach, etc. The business operations team also manages the Budget spreadsheet located in Appendix D. Further extrapolations of the budget spreadsheet are located in Appendix D.

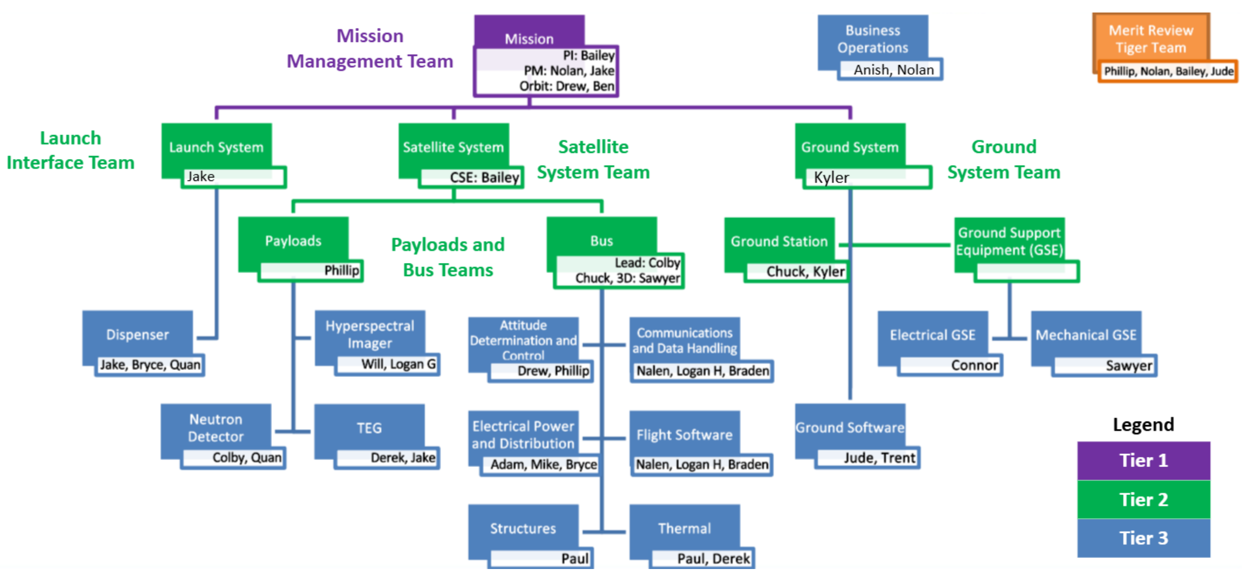


Figure 22: Organizational Structure

Risk Management

Risk management was a crucial step in the design process of this satellite. To begin this process, we began by first identifying risk. In the early stages of this project, this identification was done mainly in the form of brainstorming and research into issues that previous teams had encountered. The risks that were identified were placed by sub-teams into a collective Excel spreadsheet and scored based on likelihood and consequence. These risks were aggregated into a separate table to classify common themes. These themes were:

1. General Component Failure
2. Radiation Failure
3. Project Turnover
4. Lauch Environment
5. System Integration
6. Cost

These risks were scored according to the 2017 NASA Guidelines for Risk Management [12]. The results of that scoring can be found in the following 5x5 matrix.

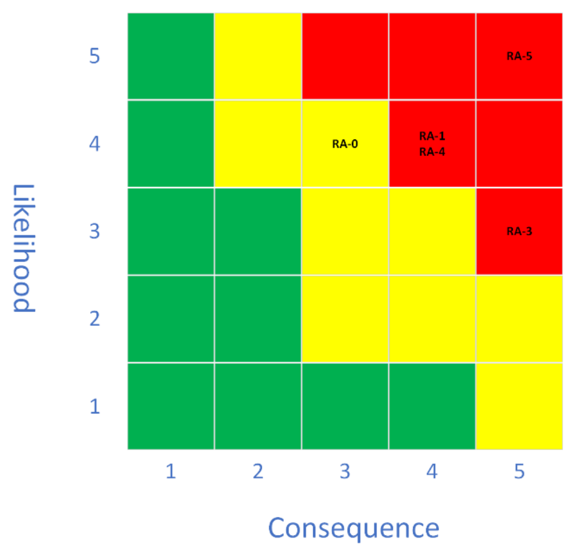


Figure 23: 5x5 Risk matrix

This mission is classified under the Class D Risk Classification [11], which is the highest risk posture accepted by NASA. Therefore, the goal of this risk mitigation is to move these risks down to at least the yellow. Moving forward, these risks will be used to make informed decisions to mitigate risk in the future.

Customer Relationship Management

For those who don't know, a customer relationship management system tracks all interactions, wants/needs, people, and information that is applicable to the project. For example, if you contact Apple Support, they have a file on you the has every issue you have ever had, what the interactions were like, what the resolution each time was, and who interacted with you. This helps steer them in the right direction when you reach out again.

For our purposes, we don't need aggressive data collection but since everyone will eventually graduate before the mission goes through it makes sense to track everything that could be applicable for the future. For example, if team member A reaches out to Garmin, the conversation and salient details should be tracked. Using the CubeSat email and date from the CRM you should be able to see the full extent of the conversation, this way you can determine the best course of action before reaching out again.

The secondary purpose of the CRM is to build a network of contacts that are aware of our project and goals. Should you need something particular from a company the CRM will quickly get you direct access to the contact list.

The tertiary purpose of the CRM is to track all Alumni in the project, as more and more students go through the program we should document personal contact information for those Alumni. These can be great resources for information and funding every academic year.

The CRM in its current form is not totally developed yet, it eventually will reach a point of where it becomes clunky on an excel sheet, when this happens and the funding allows for it, my recommendation would be to switch to a CRM platform hosted by Oracle or Monday.com or a like kind service.

The responsibility of ensuring an accurate CRM should fall under the Business Operations team as this will also help keep track of Gift-In-Kind forms and other necessary business documentation and contacts.

Merit Review Action Items

The merit review panelists had various action items that they said needed to be completed to enhance the mission and increase our chances of success. These actions items are as follows: Competitive analysis (status COMPLETE), Risk analysis (status COMPLETE), Cost viability analysis (status ONGOING), Stakeholder analysis (status ONGOING), Define mission requirements (status COMPLETE), and Knowledge transfer plan (COMPLETE)

These were the action items given as well as their status. Each analysis should be completed once a semester

Feasibility Review Action Items

The Feasibility Review panelists had action items they believe need to be addressed. These action items are finding a way to manage momentum, a way to further mitigate risk, CAD models for components, plan to manage the thermal loads, power budget, telemetry budget, fundraising plan, further address financial plan, plan for testing, and plan for redundant systems.

These were the main points from the Feasibility Review feedback. All of these need to be addressed as soon as possible.

Another request was to have more Q&A time after the presentation. This may not be possible. One option could be to present one day and have them sit on their questions until the next class period. This would split the presentation into two days.

Acronyms

ACV .... Actual Cash Value

ADC .... Attitude Determination and Control

CAD .... Computer Aided Design

CAN .... Controller Area Network

CDH .... Communications and Data Handling

CFS .... NASA's Core Flight System

CRM .... Customer Relationship Management

CS .... Computer Science

CSE .... Chief Systems Engineer

CSLI .... CubeSat Launch Initiative

CSR .... Corporate Social Responibility

CSS .... Coarse Sun Sensor

EGSE .... Electrical Ground Support Equipment

EPD .... Electrical Power and Distribution

ESRI .... Environmental Systems Research Institute

EVI .... Enhanced Vegetation Index

FOV .... Field of View

FSS .... Fine Sun Sensor

FSW .... Flight Software

GCC .... Ground Control Center

GIS .... Global Information Systems

GNSS .... Global Navigation Satellite System

GSD .... Ground Spatial Distance

GSE .... Ground Station Equipment

GYRO .... Gyroscope

IC..Integrated Circuit

KBps .... Kilobytes Per Second

Kbps .... Kilobits Per Second

LEO.. Low Earth Orbit

LNA...Low Noise Amplifier

LTAM ....Local Time of Ascending Node

MBA ....Master of Business Administration

ME .... Mechanical Engineering

MEL .... Master Equipment List

MET...Mechanical Engineering Technology

MBps .... Megabytes Per Second

Mbps .... Megabits Per Second

MOC ....Missions Operation Center

MTM..Magnetometer

MTQ ....Magnetorquer

NDVI ....Normalized Difference Vegetation Index

NE .... Nuclear Engineering

NASA .... National Aeronautics and Space Administration

OBC .... On-board Computer

PCDU...Power Conditioning and Distribution Unit

PHAT-Sat .... Precision Hyperspectral Agricultural Tracking Satellite

PI .... Principal Investigator

PM ....Principal Manager

PLD...Payload

PV..Photovoltaic

Q&A .... Questions and Answers

RDT .... Radiation Detection Technologies

RF .... Radio Frequency

RS-422 .... Recommended Standard 422

RWL .... Reaction Wheel

S3 .... Amazon Simple Storage Service

SDR .... Software Defined Radio

SKU .... Stock Keeping Unit

SME .... Subject Matter Expert

SSH .... Secure Shell

STR .... Structure

TNC .... Terminal Node Controller

TTL .... Time To Live

TEG .... Thermoelectric Generator

UART .... Universal Asynchronous Receiver-Transmitter

UHF .... Ultra-High Frequency

UML .... Unified Modeling Language

USB .... Universal Serial Bus

VHF .... Very High Frequency

VNIR .... Visible Near-Infrared

VPN .... Virtual Private Network

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Appendices

Appendix A: Requirements Verification Matrix

PHAT-Sat Requirements Verification Matrix.xlsx

Appendix B: Master Equipment List Spreadsheet

PHAT-Sat Master Equipment List Spreadsheet

Appendix C: Software Development Plan

Introduction

The Flight Software, referred as FSW, will be a complex project with a high turnover team. It will be of utmost importance to remain organized and to always develop with this plan in mind.

This plan will outline the framework chosen and how the FSW development project will be organized, specifically what roles will be necessary and the responsibilities of each role. Next, it will outline the schedule and it will also discuss all the foreseen risks that need to be mitigated. Furthermore, the methodology for managing the project will be discussed, and how the quality will be assured throughout the development. Finally, the configurations that will be put in place will be outlined, as well as the documents that are needed to be produced and the way that the project will be monitored until completion.

Framework

For this project, NASA's core Flight System will be utilized. This framework has been proven in over 40 missions so far on a wide variety, including CubeSats. According to NASA's website, it is a "platform-independent, reusable software framework designed to expedite flight software development" [12].

This framework is built by NASA engineers to be able to share the expertise they possess, and allow for a simpler software development process for engineers working on Flight Software [12]. Furthermore, it allows new projects to avoid starting from the very beginning, and also helps ensure quality and risk management as it has been tested many times on real missions [12].

Finally, another contributing factor for selecting this framework was the expansive training resources that NASA provides for it. NASA provides an FSW 101 course as well as a cFS 101 course to prepare engineers for utilizing this framework.

The full description of what NASA core Flight System framework entails can be found here: https://etd.gsfc.nasa.gov/capabilities/core-flight-system/ [12].

Project Organization

The specific personnel for each specific responsibility are yet to be determined, but the necessary roles that need to be filled to ensure successful development are listed below.

Flight Software Lead – This role will be responsible for managing the FSW development program. They will oversee the timeline, budget, and communicate with other subsystem teams. They will additionally take ownership of the code repository and ensure that FSW risks are mitigated.

Embedded Systems Engineer – Develops low-level software such as device drivers and interfaces. Responsible for working closely with the system hardware and is responsible for the integration of the hardware (sensors, microcontrollers, etc.) with the high-level software.

Payload Software Integration Engineer – Develops software that will interface with the mission payload (hyperspectral imager, radiation detector). This role will specialize in the control, timing, and management of data collected from payload. Additionally, they will create a software solution for compressing data in preparation from downlink handled by the CDH team.

Communications and Data Handling Engineer – Works with CDH team to integrate the CDH systems with the FSW.

Ground Station Software Liaison Engineer – Works with Ground Station team to ensure successful compatibility and ensures consistency between FSW and Ground Station Software

Project Schedule

Specific dates for deadlines and milestones are currently unknown, and milestones will be set as needed. These milestones will correlate to Sprints (refer to Development Methodology section).

Risk Management

Risk: Memory Corruption

Mitigation: Ensure that sufficient error checking and memory-safe design patterns are applied. This includes avoiding any and all dynamic memory allocation, bounded recursion if any, and adherence to other principles outlined in NASA's *The Power of 10: Rules for Developing Safety-Critical Code* [13]. If we can adhere to these guidelines, then this risk will be mitigated.

Risk: Rushed Timeline

Mitigation: In general, this will be a risk that we will have to accept. The best way that we will be able to mitigate this is through early planning and adherence to deadlines. This will allow us to ensure that we don't fall behind and need to "make up the time elsewhere". Using task trackers like Trello will be important to ensure that everyone on the team is on the same page with development processes. Additionally, we should also plan sufficient time for testing our code. While fast development is important, the primary focus should be on ensuring that our code is functional and safe. If we can adhere to these guidelines, then this risk will be mitigated.

Risk: Unhandled errors

Mitigation: Identify potential failure points in our code early during design reviews. In addition to this we should have multiple levels of error handling to help ensure a logical control flow during satellite operation. This will allow for uninterrupted code execution and system stability throughout different operational modes and ground station commands. Additionally, all code should pass before multiple sets of eyes. This will help reduce errors that a single person may have missed. If we can adhere to these guidelines, then this risk will be mitigated.

Development Methodology

This software development process will utilize the Scrum framework which is part of Agile software development methodology.

Scrum consists of, according to the official Scrum documentation, "increments of valuable work are delivered in short cycles of one month or less, which are called Sprints. Ongoing feedback occurs during the Sprint, allowing for inspection and adaptation of the process and what will be delivered" [14]. There are many roles on a Scrum team like the Scrum Master, Product Owner, and Developers that contribute to the project [14].

The full description of what the Scrum framework entails can be found here: www.scrum.org [16].

Quality Assurance

For this project we will have multiple levels of quality assurance designed to ensure that our code is both functional and safe. At the top level we will employ requirements-based design, and consequentially requirements-based testing. Requirements-based design is the idea that all functional code should be for the purpose of fulfilling mission-driving requirements. Requirements-based testing is the idea that all tests should be derived directly from the requirements of the system. Enacting requirements-based design and requirements-based testing will ensure that we result in a codebase that is thoroughly tested and proven to work.

With this project we will have a strict process workflow with integrating functional code into our codebase. Once functional code is written it will first go through multiple rounds of review. This will involve requiring justifications for any new code and modified code as well as highlighting what requirements these additions flow down from. Lastly, before a final shipment of the FSW system, all functional code will go through Mod testing to ensure that each module functions as intended and without error.

All functional code should also have multiple levels of error-handling. This will ensure uninterrupted code execution throughout the satellite's lifetime. Additionally, there should be no instances of deadlock or non-reentrant code.

Furthermore, all functional code should follow NASA's *The Power of 10: Rules for Developing Safety-Critical Code* [13]. This will give us strong guidelines that focus on safe code. The rules developed in this paper are as follows:

1. Restrict all code to very simple control flow constructs – do not use goto statements, setjmp or longjmp constructs, and direct or indirect recursion.
2. All loops must have a fixed upper-bound.
3. Do not use dynamic memory allocation after initialization.
4. No function should be longer than what can be printed on a single sheet of paper in a standard reference format with one line per statement and one line per declaration. Typically, this means no more than about 60 lines of code per function.
5. The assertion density of the code should average to a minimum of two assertions per function.
6. Data objects must be declared at the smallest possible level of scope.
7. The return value of non-void functions must be checked by each calling function, and the validity of parameters must be checked inside each function.
8. The use of the preprocessor must be limited to the inclusion of header files and simple macro definitions.
9. The use of pointers should be restricted. Specifically, no more than one level of dereferencing is allowed. Pointer dereference operations may not be hidden in macro definitions or inside typedef declarations. Function pointers are not permitted.
10. All code must be compiled, from the first day of development, with all compiler warnings enabled at the compiler's most pedantic setting. All code must compile with these setting without any warnings.

By following these rules, we will ensure that we are writing quality code, and that this code is thoroughly checked and safe.

Configuration Management

The version control system used in this development process will be Git, and the repository platform GitHub will be used to store and manage code files. The Flight Software Lead will have control of the repository, as well as final say in commits.

To commit changes to the repository, developers will submit a Pull Request for review by the Flight Software Lead. Then, it will either be accepted into the repo, or it will be declined and a request for updates will be sent to the developer who will thing submit a new Pull Request once the updates are complete.

Documents

In this process, documentation will be of utmost importance due to the high turnover of the software development team. As a new team of students will begin working on this each semester of the project, it will be important to document code written and processes followed.

The following documents shall be produced over the course of the development process:

Technical Document – This document will outline all technical information like languages and libraries used, structure of file system, interfacing details with other subsystems for the satellite, and more. This document should allow someone to understand the details and methodologies for each component of the software.

Transition Document – This document will provide information in addition to the Technical Document that will help with smooth transitions between students entering and exiting the team. It will contain information like lessons learned, ideas for future process improvement, important contacts, and the contact information for each member who has worked on the project and how they contributed.

UML Design – This document will contain the UML (Unified Modeling Language) diagram for the software. This shows each class, the attributes and methods of each, as well as how each class connects to create the overall functionality of the software.

Project Monitoring and Control

Each member of the development team shall keep open and frequent communication, and keep each other member of the team up-to-date on what each member is currently working on. The Flight Software Lead will monitor overall progress towards software completion as well as plan Sprints. At least once a week, the development team shall meet for a stand-up to share progress and challenges in development.

Appendix D: Budget Spreadsheet

There are two different sheets that govern the budget spreadsheet, the first sheet is the Balance sheet. This tracks the standing Actual Cash Value (ACV) in the account at any given time, It is the master sheet which should be used to compare the Concur and Foundation reports against for increased accuracy. The second sheet is the expense sheet. It exists to purely track expenses by subsystem and groups to rapidly understand where we are in a spend cycle (spending is cyclical in most design projects).

The sheets are different as they should eventually have different levels of access, like the expense sheet should be available for all to see, but the balance sheet should be restrictive.

Table 11: Balance Sheet Headers

Type	System	Company	Income	Expense	Sponsorable?	KSU-POC	KSU-POC	KSU-Advisor	Date	Income	$0.00	Expense	$0.00	Delta	$0.00

These are the headers from the balance sheet, for any income or expense that is recorded, two students and an advisor should put their initials in the box to confirm validity of the transaction. This prevents fraud, double orders, and forces more people to be in the loop without bringing too many people into the loop.

Table 12: Expense Sheet

	Total Cost PHAT-SAT								
Mission	$0.00								
Satellite	$0.00								
Launch	$0.00								
Ground	$0.00								
Total	$0.00								
Subteam	Item	Company	QTY	Total Cost	Lead Time	Ordered	Receipt	Purcahsed By	Notes
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		
						FALSE	FALSE		

The expense sheet on the other hand includes realized expenses and projected expenses which is why this information does not need to be restrictive in nature. This gives everyone a clear understanding of what is happening financially at 30K foot view.

It also lets people rapidly find quotes and old expenses down to the SKU number by having receipts linked in. The lead time category for this design project is critical due to the potential long lead times and short turnarounds that are needed.

Appendix E: Schedule Spreadsheet

The Schedule spreadsheet is in the Teams general files. There are 6 columns that will be filled out per task. Whoever has the task of managing the schedule will be in charge of filling out 4 of these columns. The other two columns will be filled out by the PI and Professor after the task has been completed. This ensures that there is a 3 tier sign off system to make sure nothing gets accidentally checked off. The schedule can be added to or removed from with the collective agreement between the project management group and the professor.

Table 13: Schedule Spreadsheet

B	C	D	E	F	G	H
Assigned Semester	***Start Date***	***Semester Completed***	**P.I. Sign Off**	**Professer Sign Off**	**Scheduler Sign Off**	**Has Everyone Signed Off?**
Spring 2025	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Fall 2025						NO
						NO
						NO
						NO
						NO
						NO
						NO
						NO
						NO
						NO

This is how the schedule will look when complete versus incomplete.

Table 14: Full Schedule

Milestones	Assigned Semester	Start Date	Semester Completed	P.I. Sign Off	Professer Sign Off	Scheduler Sign Off	Has Everyone Signed Off?
System Engineering Phase	Spring 2025	N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Mission Formulation/Merit review		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Requirements Review		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Conceptual Design/Feasibility Review		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Design showcase		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Final Report for Semester Data Transfer		N/A	Spring 2025	Bailey Walke	Paul Snider	Nolan Elder	YES
Preliminary Design Proposal Phase	Fall 2025						NO
Previous Semester Report Review							NO
Submit CSLI Proposal							NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation							NO
Schedule Review							NO
Risk Reevaluation							NO
Preliminary Design/Pre-PDR Review							NO
Software Developement							NO
Final Report for Semester Data Transfer							NO
Detailed Design Phase	Spring 2026						NO
Previous Semester Report Review							NO
Preliminary/Critical Design and Reviews							NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation							NO
Schedule Review							NO
Risk Reevaluation							NO
CSLI Selection							NO
Procure Hardware (pending CSLI selection)							NO
Design showcase							NO
Final Report for Semester Data Transfer							NO
AIT Phase	Fall 2026 - Spring 2028						NO
Previous Semester Report							NO
Build							NO
Test							NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation							NO
Schedule Review							NO
Risk Reevaluation							NO
Resubmit CSLI Proposal (as needed)							NO
Design showcase							NO
Final Report for Semester Data Transfer							NO
Launch	Fall 2028						NO
Previous Semester Report Review							NO
Mid Semester Progress Review							NO
Stakeholder/Budget Review							NO
Mission Advisory Board Progress Review							NO
Licensing Evaluation							NO
Schedule Review							NO
Risk Reevaluation							NO
Pending Launch Manifest							NO
Final Compiled Mission Report							NO

Appendix F: Customer Relationship Management spreadsheet

Table 15: Customer Relationship Spreadsheet

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Company	Status	System	KSU-Primary Contac	Primary Contact	Email	Phone Numbe	ACV	Last Contact	Type of Aid	Category	Thank You	GIK	Mailing Address	Notes
											FALSE	FALSE		
											FALSE	FALSE		
											FALSE	FALSE		
											FALSE	FALSE		
											FALSE	FALSE		
											FALSE	FALSE		

The CRM tracks many categories, from point of contacts to active status to follow up tasks like Thank You notes and Gift-In-Kind forms. Understanding and filling out every single box is crucial to maintaining proper documentation and knowledge transfer. Most companies budget annual contributions in their philanthropy budget, utilizing different companies CSR (Corporate Social Responsibility) policies you may be able to get more monetary and in-kind value from them.

The CRM should eventually be locked down to only subsystem leads and higher to preserve the integrity of the information within the sheet and to safeguard sponsor and donor information in the best way possible.

Appendix G: SolidWorks Renderings

Preface

The general dimensions of the satellite along with the driving envelope requirements from Nanoracks are given first. After these two, numerous different images of the satellite are provided for reference and any future use. If any other images are needed, consult the CAD file. Additionally, a motion study animation rendering was performed in SolidWorks that displays the internal arrangement – this is found, again, in the CAD files, but specifically within the Design Showcase folder.

Images

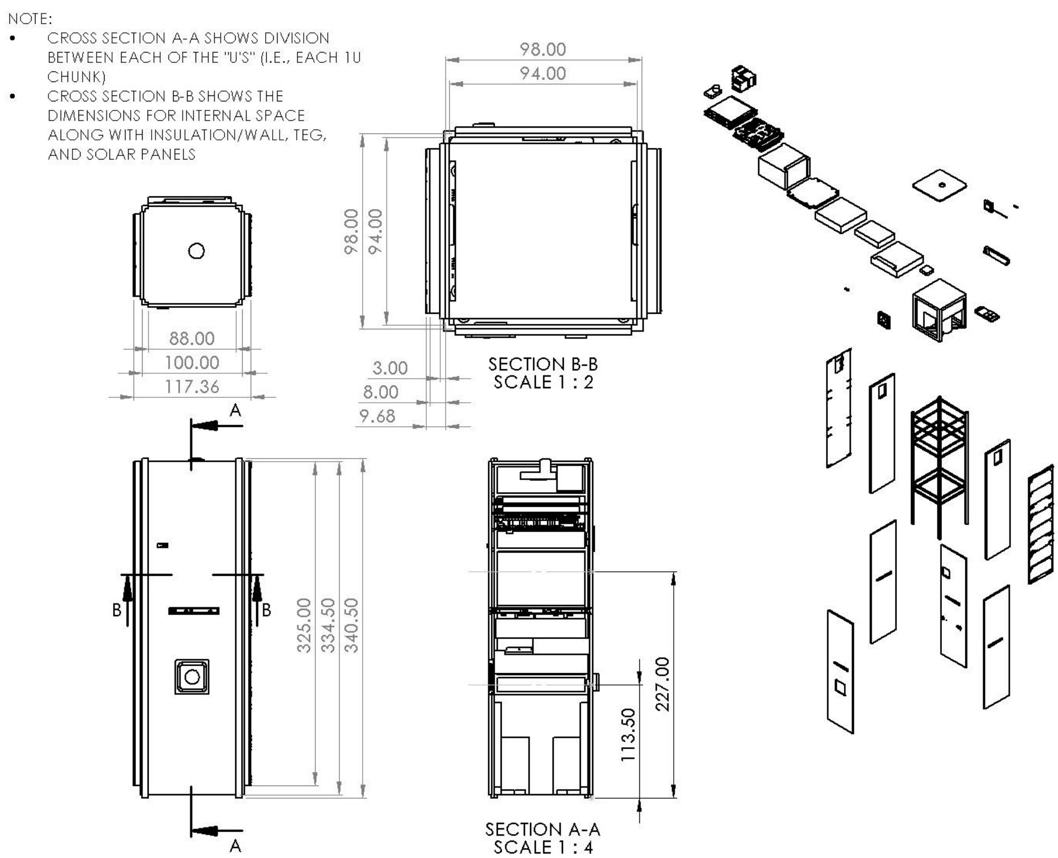

Figure 24: Overall Satellite Dimensions (REV 1 Layout)

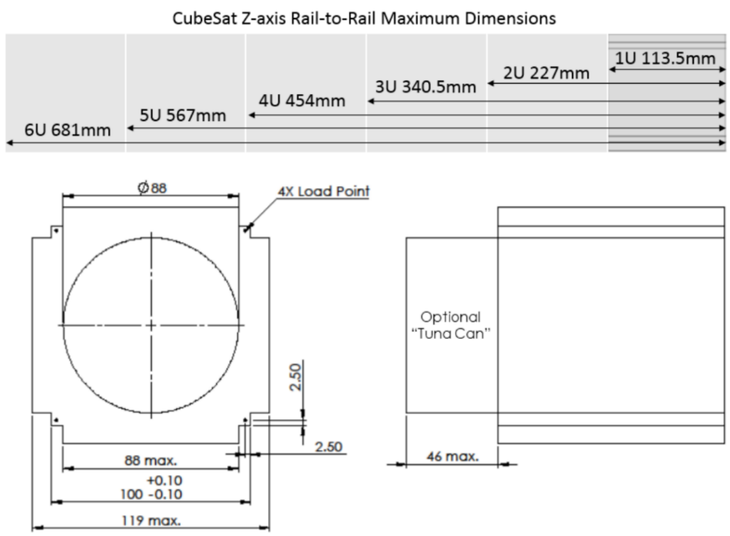

Figure 25: Nanoracks 2018 Envelope Compliance Reference

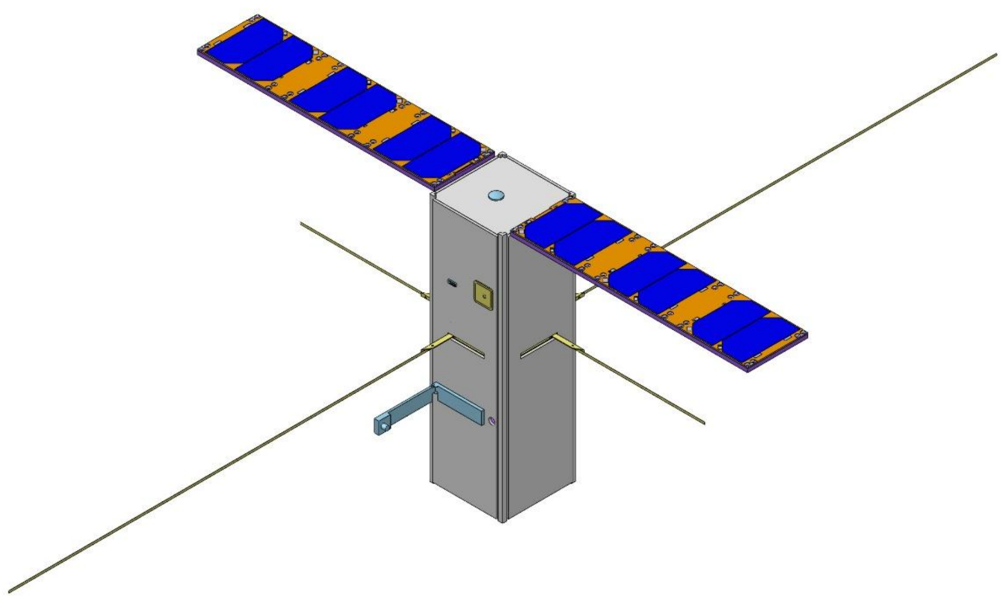

Figure 26: Isometric View (Back, Color, Extended)

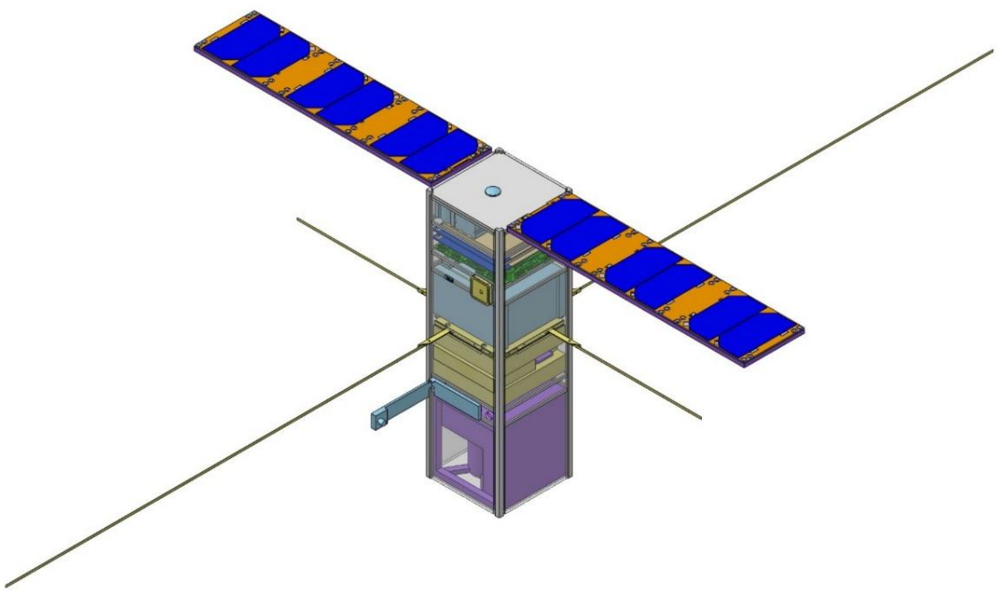

Figure 27: Isometric View (Back, Color, Extended, Transparent)

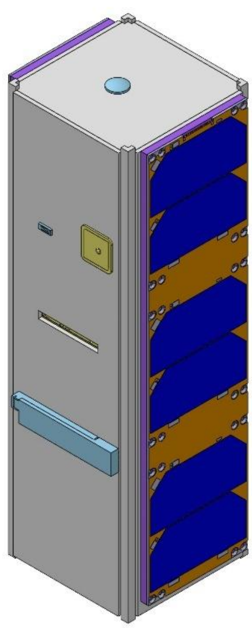

Figure 28: Isometric View (Back, Color, Stowed)

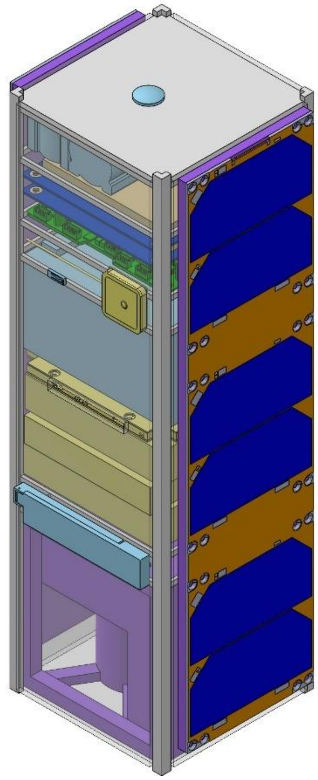

Figure 29: Isometric View (Back, Color, Stowed, Transparent)

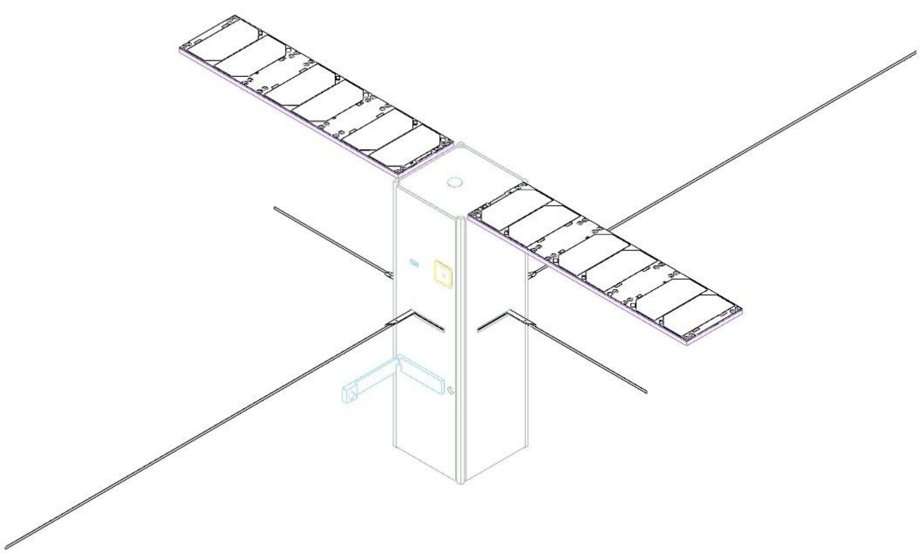

Figure 30: Isometric View (Back, Wire, Extended)

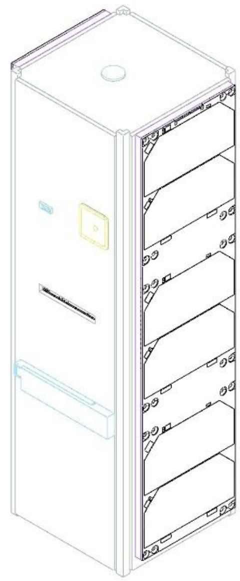

Figure 31: Isometric View (Back, Wire, Stowed)

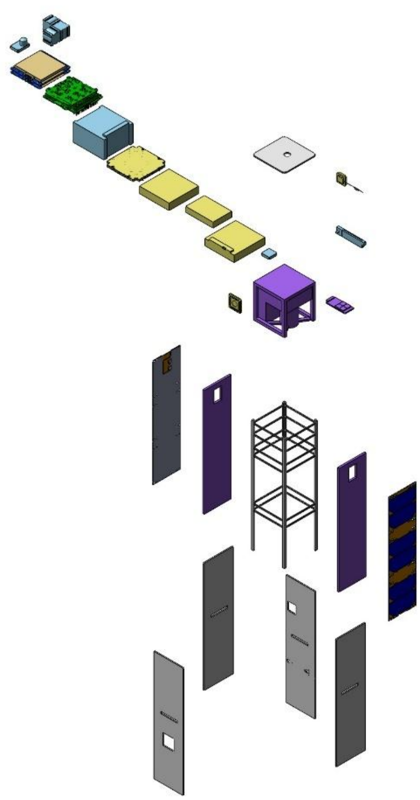

Figure 32: Isometric View (Exploded-All)

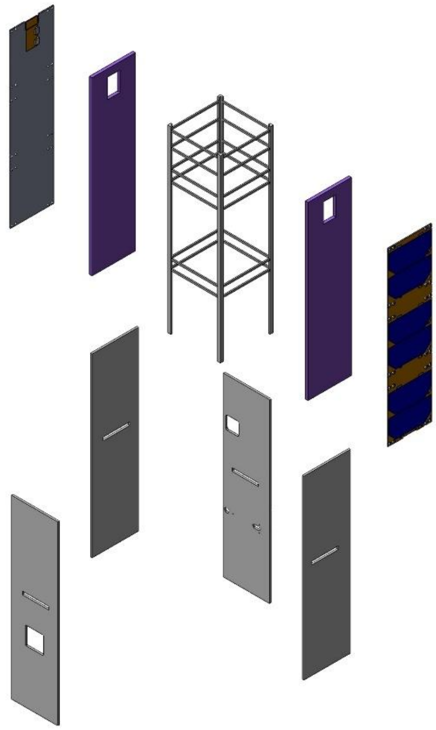

Figure 33: Isometric View (Exploded-Externals)

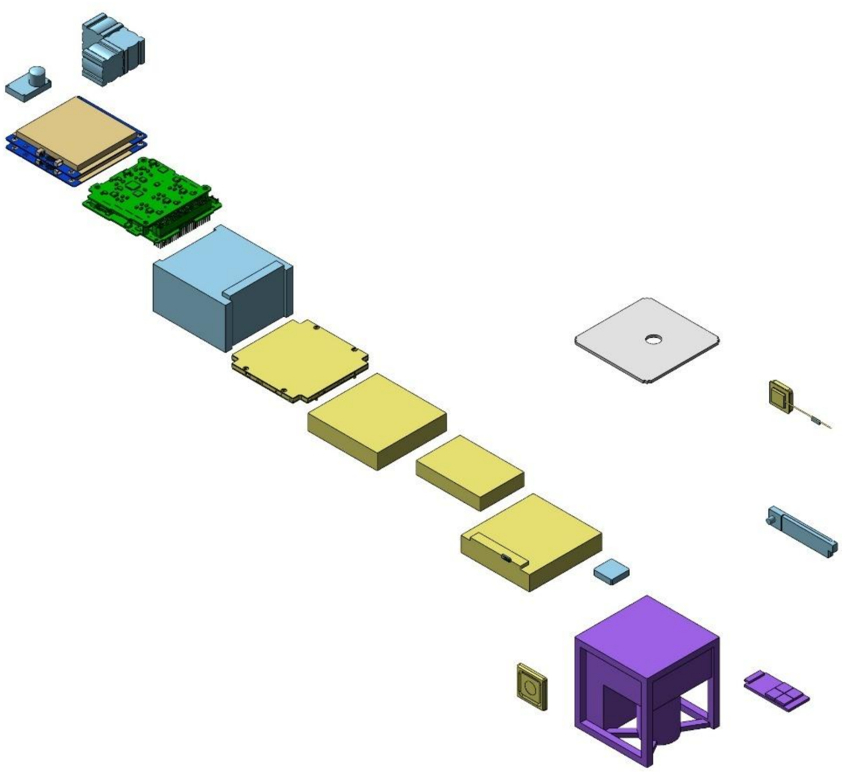

Figure 34: Isometric View (Exploded-Internals)

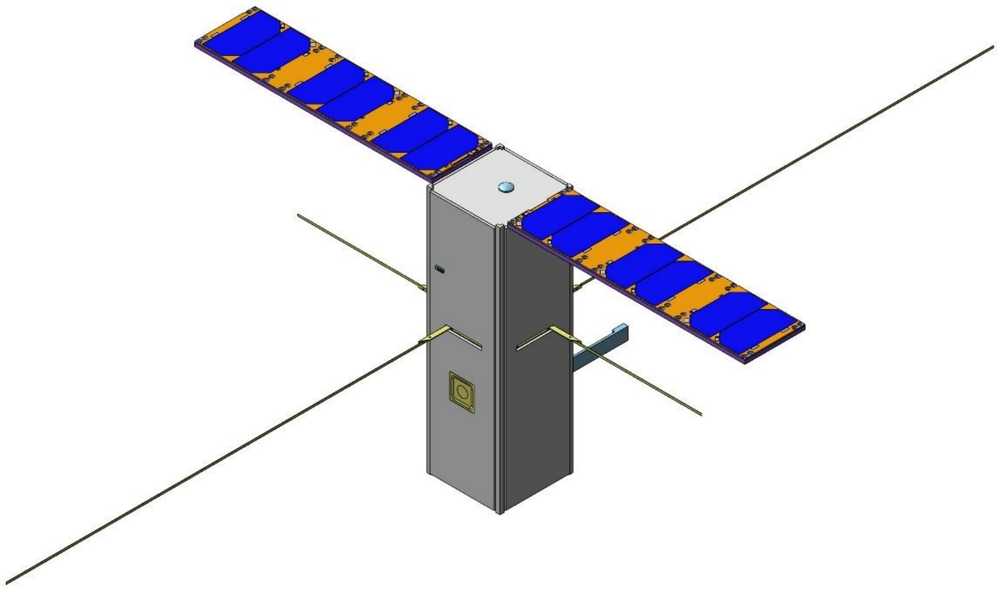

Figure 35: Isometric View (Front, Color, Extended)

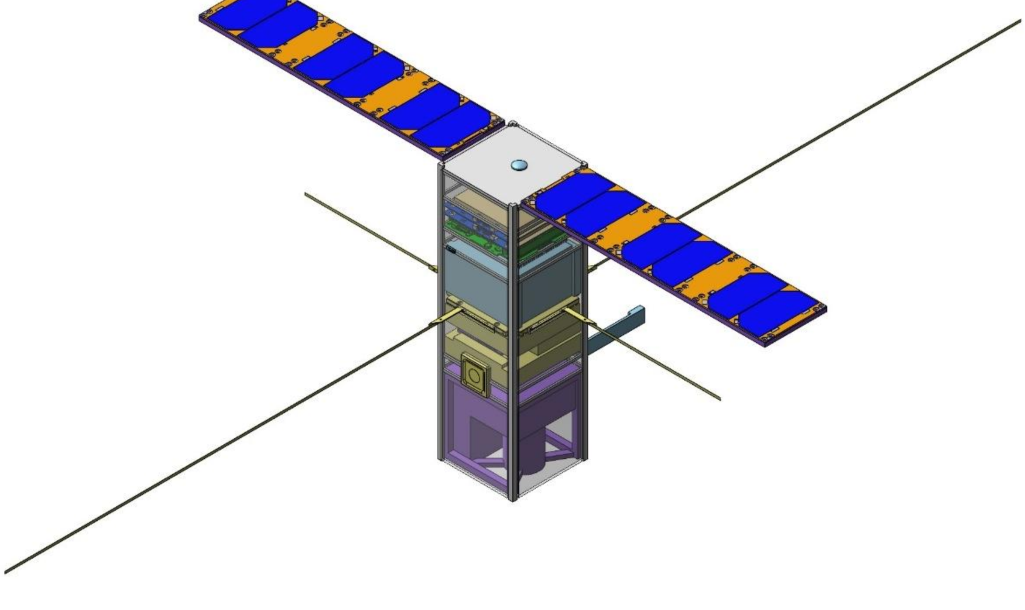

Figure 36: Isometric View (Front, Color, Extended, Transparent)

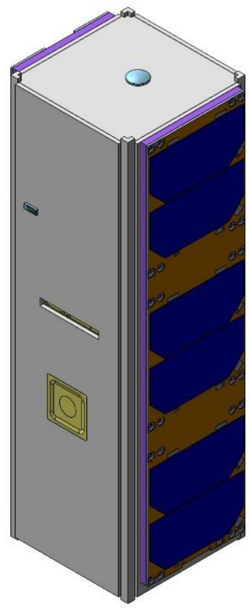

Figure 37: Isometric View (Front, Color, Stowed)

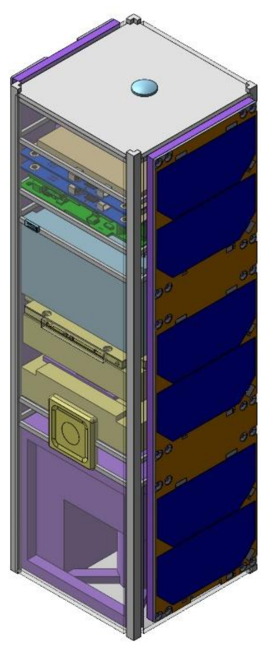

Figure 38: Isometric View (Front, Color, Stowed, Transparent)

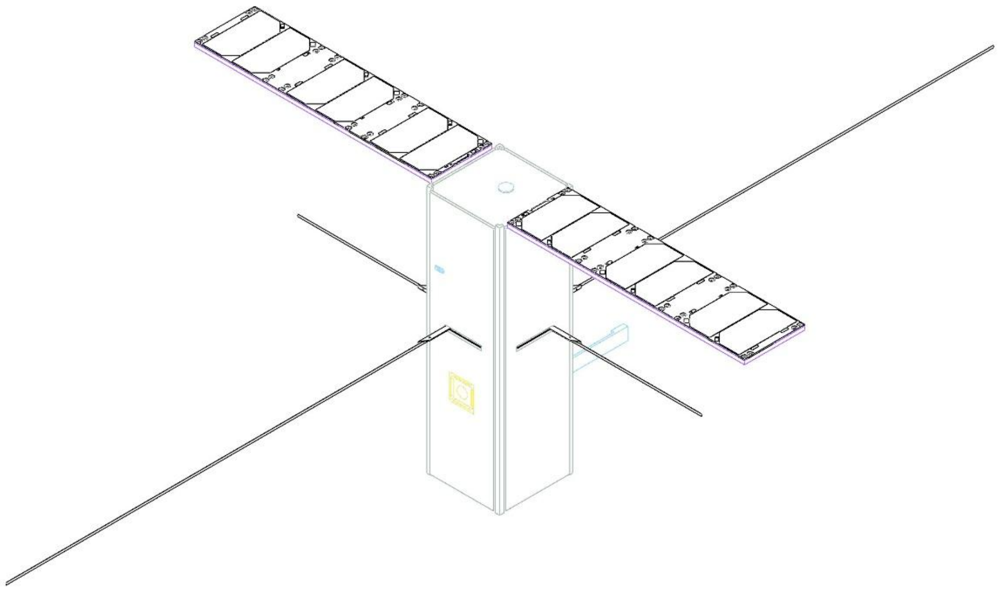

Figure 39: Isometric View (Front, Wire, Extended)

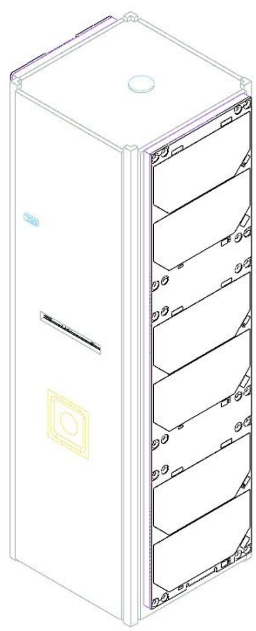

Figure 40: Isometric View (Front, Wire, Stowed)